Assignment 2

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Introduction

Task 1 section details the methodology and results for forecasting monthly average horizontal solar radiation two years ahead (January 2015 to December 2016). The analysis utilized the solar radiation series itself alongside monthly precipitation as an exogenous variable over the period spanning January 1960 to December 2014. To ensure the robustness and accuracy of the forecast, four distinct classes of time series models were evaluated: time series regression, dynamic linear models, and exponential smoothing models, and corresponding state-space models. The goal was to identify the model producing the lowest Mean Absolute Scaled Error (MASE), providing the best 24-month forecast.

Task 2 investigates the relationship between two critical economic indicators in Victoria: the quarterly Residential Property Price Index (PPI) in Melbourne and the quarterly population change (transformed logarithmically) between September 2003 and December 2016. The primary objective of this report is to determine if this high correlation is genuine or if it is spurious, resulting purely from shared non-stationary trends and seasonality within the data, and, consequently, to identify the true short-run relationship between their growth rates.

Task 1: Analysis and forecasting

```r
#loading packages
library(dLagM)
library(dynlm)
library(forecast)
library(ggplot2)
library(tseries)
library(TSA)
library(Hmisc)
library(xts)
library(AER)
library(urca)
library(tseries)
library(dynlm)
```

Data Description

- The raw historical and future precipitation data were imported and validated, with the columns being renamed for clarity. All variables were then converted into R time series objects (ts), correctly defined as monthly data with a

frequency of 12, beginning in January 1960, to ensure proper handling of seasonality during model estimation.

```
#read data
setwd("C:/Users/tally/OneDrive/Desktop/Forecasting/Assignment2")
solar_data <- read.csv("data1.csv")
#check data
head(solar_data)
```

```
##       solar   ppt
## 1  5.051729 1.333
## 2  6.415832 0.921
## 3 10.847920 0.947
## 4 16.930264 0.615
## 5 24.030797 0.544
## 6 26.298202 0.703
```

```
#rename column
colnames(solar_data)<-c("solar", "precipitation")
future_precip <- read.csv("data.x.csv")
head(future_precip)
```

```
##           x
## 1 0.1890100
## 2 0.6972625
## 3 0.5952135
## 4 0.4873885
## 5 0.2616770
## 6 0.8086067
```

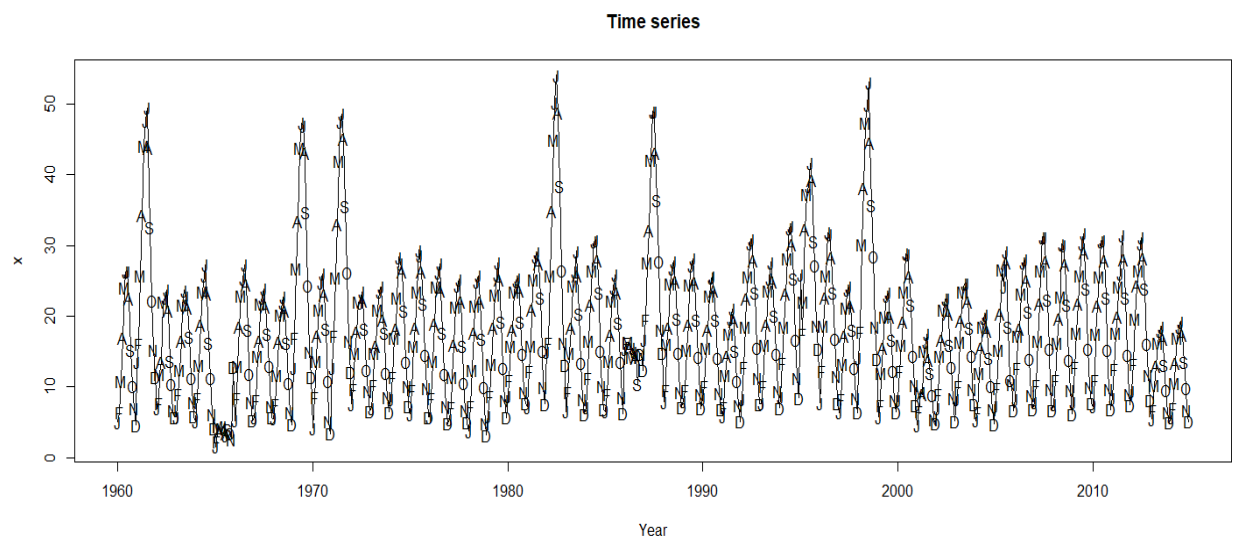
```
#Time series object
solar_ts <- ts(solar_data$solar, start = c(1960, 1), frequency = 12)
precip_ts <- ts(solar_data$precipitation, start = c(1960, 1), frequency = 12)
data_ts <- ts(solar_data, start = c(1960, 1), frequency = 12)
future_precip_ts <- ts(future_precip, start = c(2015, 1), frequency = 12)
```

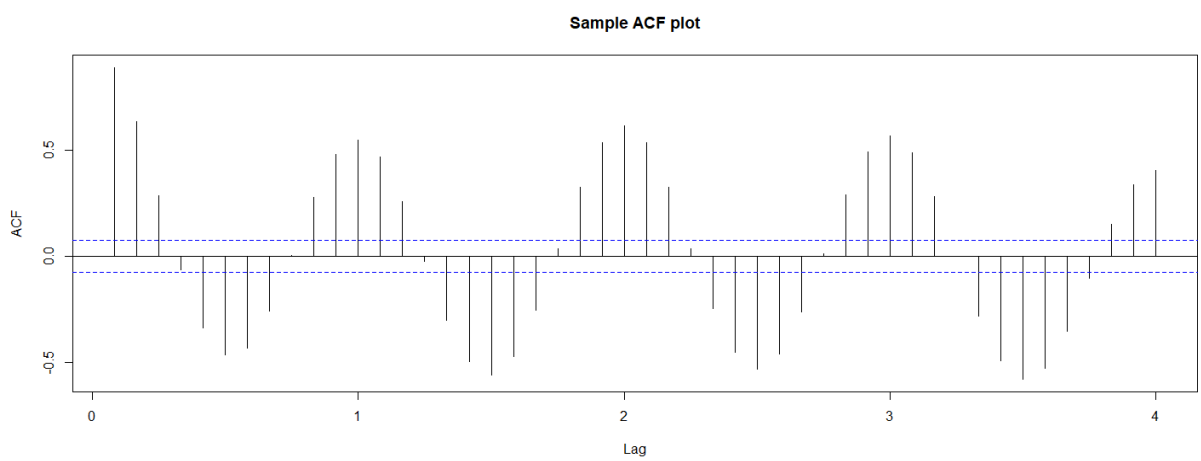
Data Exploration

```
plots <- function(x) { #function to explore time series objects: time series
plot, ACF plot, ADF test and PP test
  plot(x, xlab='Year', main="Time series")
  points(y=x,x=time(x), pch=as.vector(season(x)))
  acf(x, main="Sample ACF plot", lag.max = 48) #ACF plot
  adf_result <- ur.df(x, type = "none", lags = 1, selectlags = "AIC") #ADF
test
  print(summary(adf_result))
  pp_result <- ur.pp(x, type = "Z-tau", lags = "short") #PP test
  print(summary(pp_result))
}
```

Solar radiation object

- The data exhibits strong, consistent annual seasonality, peaking around July and reaching troughs around December. The magnitude of these fluctuations appears to vary over time.
- Both ADF and PP tests (test statistic is less tan all critical values) indicate that the series is stationary (the null hypothesis of a unit root is rejected) which means the mean and variance of the series are stable over time, and no regular differencing is required for modeling





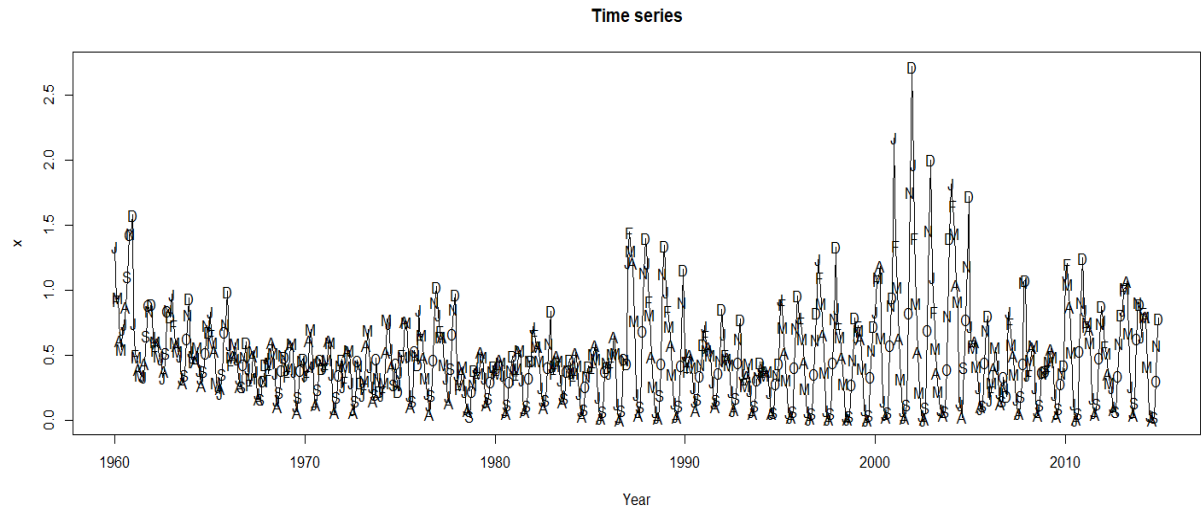

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
```

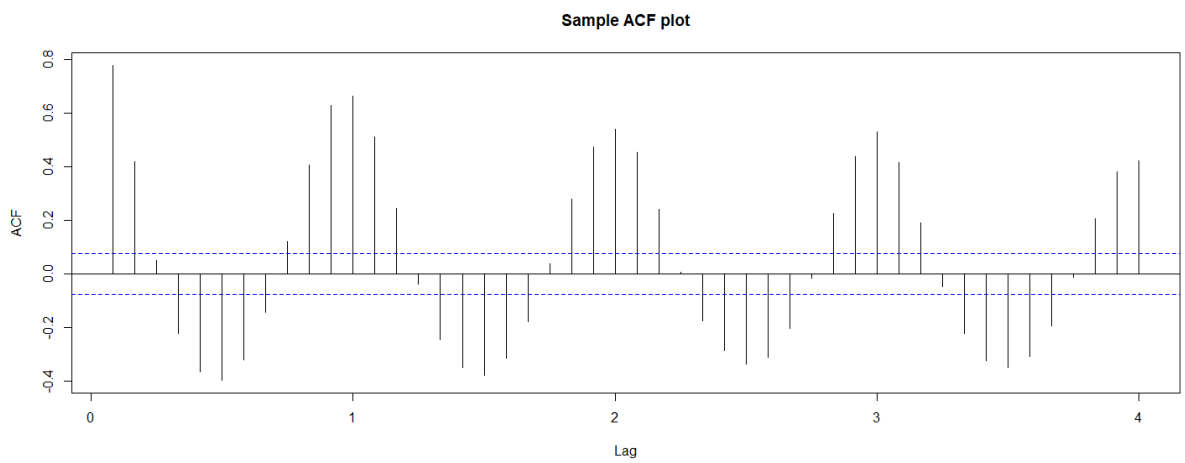
```
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4027  -1.3527   0.5471   2.6802  19.7515
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## z.lag.1    -0.042790   0.006363  -6.725 3.82e-11 ***
## z.diff.lag  0.689323   0.028286  24.369  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.299 on 656 degrees of freedom
## Multiple R-squared:  0.4818, Adjusted R-squared:  0.4802
## F-statistic:   305 on 2 and 656 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: -6.7251
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
##
##
## ##################################
## # Phillips-Perron Unit Root Test #
## ##################################
##
## Test regression with intercept
##
##
## Call:
## lm(formula = y ~ y.l1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.6478 -3.7383  0.3732  3.3766 16.4265
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.9384     0.3598   5.388 9.92e-08 ***
## y.l1          0.8911     0.0177  50.350  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.453 on 657 degrees of freedom
## Multiple R-squared:  0.7942, Adjusted R-squared:  0.7939
## F-statistic:  2535 on 1 and 657 DF,  p-value: < 2.2e-16
##
```

```
##
## Value of test-statistic, type: Z-tau  is: -8.4301
##
##          aux. Z statistics
## Z-tau-mu           7.3858
##
## Critical values for Z statistics:
##                      1pct      5pct     10pct
## critical values -3.442671 -2.866274 -2.569292
```

Precipitation object

- The time series plot shows there is a slight increasing trend in its average level and volatility. The series describes strong seasonality, and a consistent pattern of peaks and troughs appears to repeat within each year.
- The ACF plot with the strongest positive correlations is clearly visible around the 1-year, 2-year, 3-year, and 4-year marks, confirming a strong annual seasonality
- The ADF and PP test shows the same result as the solar object when both of them have a test statistic that is smaller than the critical value. Therefore, the test indicates the series is stationary.
- To conclude, the time series is stationary but exhibits a strong annual seasonal pattern and high short-term persistence





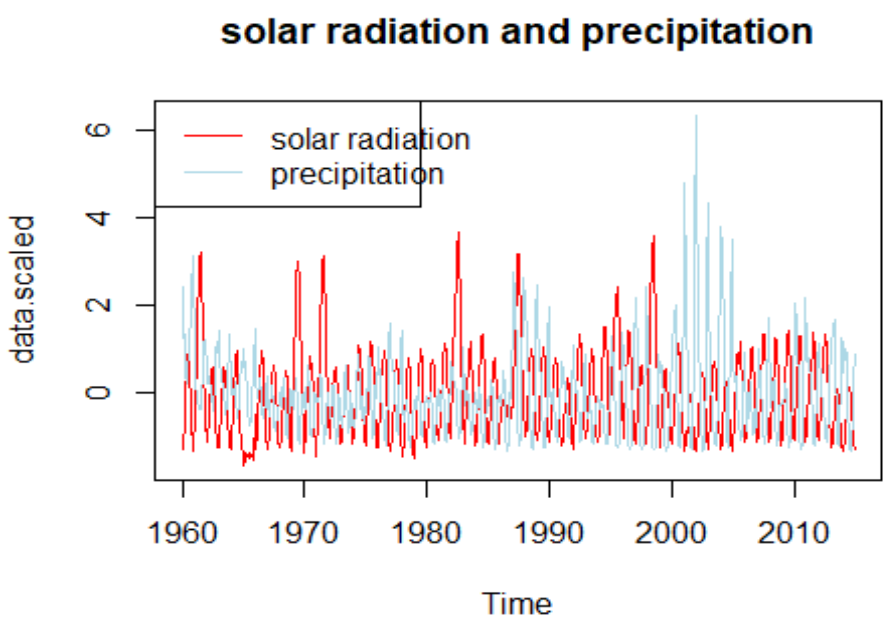

```
##
## ################################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ################################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.05939 -0.04034  0.04809  0.14327  1.30427
##
## Coefficients:
##            Estimate Std. Error t value Pr(>|t|)
## z.lag.1    -0.10907    0.01434  -7.607 9.78e-14 ***
## z.diff.lag  0.38211    0.03593  10.635  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2113 on 656 degrees of freedom
## Multiple R-squared:  0.1813, Adjusted R-squared:  0.1788
## F-statistic: 72.65 on 2 and 656 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: -7.6069
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
##
##
## ###################################
## # Phillips-Perron Unit Root Test #
## ###################################
##
## Test regression with intercept
##
##
## Call:
## lm(formula = y ~ y.l1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.71989 -0.12860 -0.02614  0.09731  1.33078
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10248    0.01422   7.209 1.55e-12 ***
## y.l1         0.77928    0.02420  32.195  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2204 on 657 degrees of freedom
## Multiple R-squared:  0.6121, Adjusted R-squared:  0.6115
## F-statistic:  1037 on 1 and 657 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic, type: Z-tau  is: -9.3544
##
##          aux. Z statistics
## Z-tau-mu             7.399
##
## Critical values for Z statistics:
##                      1pct      5pct     10pct
## critical values -3.442671 -2.866274 -2.569292
```

Correlation examination

- Before fitting the model, examining the relationship between two objects will bring a helpful vision of the correlation between the explanatory variable and the dependent variable
- Across the entire time horizon, there is a clear inverse seasonal correlation between the two variables. Periods of high solar radiation generally coincide with periods of low precipitation, and vice versa.
- The correlation coefficient shows a negative value of -0.45, showing a moderate relationship between these variables.

```
data.scaled <- scale(data_ts) # scale data to the same measure
plot(data.scaled, plot.type="s", col=c("red", "lightblue"), main = "solar
radiation and precipitation")
legend("topleft", lty=1, text.width = 11, col = c("red", "lightblue"),
c("solar radiation", "precipitation"))
```




```
cor(data_ts) #correlation coefficient

##                     solar precipitation
## solar          1.0000000    -0.4540277
## precipitation -0.4540277     1.0000000
```

I. Fitting Models

1. Time Series Regression Methods

a. Finite DLM

- The loop shows that the lowest MASE belongs to lag 12, so we will execute the finite DLM with lag 12

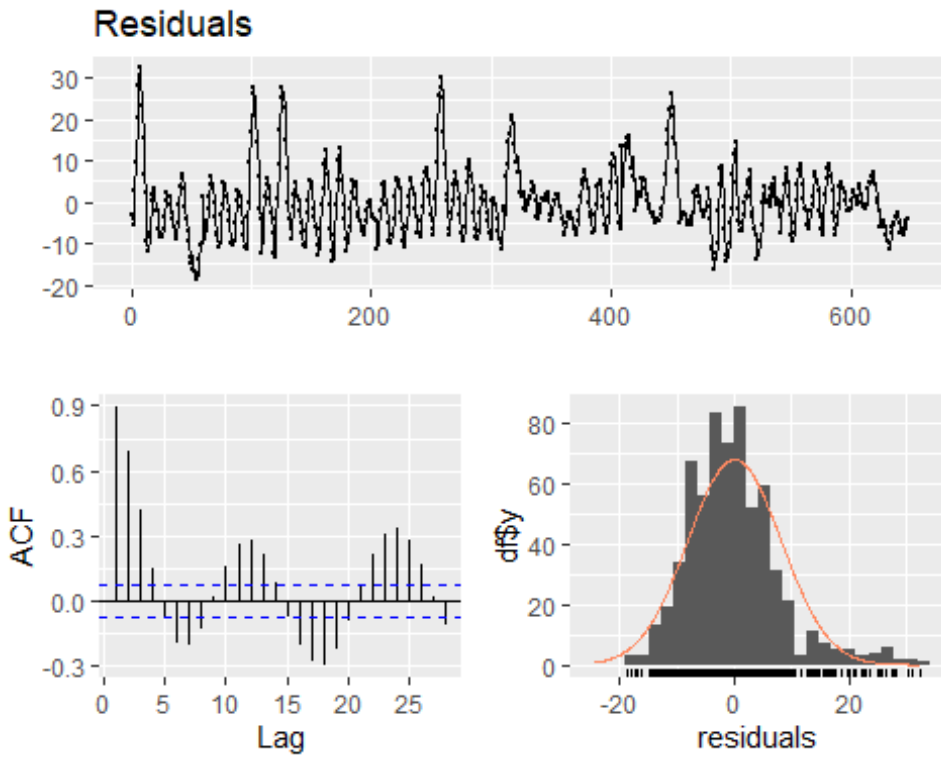
```
summarise <- function(x) { #function to summarise models
  summary(x)
  checkresiduals(x$model)
  VIF <- vif(x$model)
  VIF >10
}
# loop for identifying number of lag for model
for (i in 1:12) {
  model1.1 <- dlm(x = as.vector(solar_data$precipitation),
                  y = as.vector(solar_data$solar),
                  q = i)
  cat("q =", i,
      "AIC =", AIC(model1.1$model),
      "BIC =", BIC(model1.1$model),
      "MASE =", MASE(model1.1)$MASE,
      "\n")
}
```

```
## q = 1 AIC = 4728.713 BIC = 4746.676 MASE = 1.688457
## q = 2 AIC = 4712.649 BIC = 4735.095 MASE = 1.675967
## q = 3 AIC = 4688.551 BIC = 4715.478 MASE = 1.662703
## q = 4 AIC = 4663.6 BIC = 4695.003 MASE = 1.646357
## q = 5 AIC = 4644.622 BIC = 4680.499 MASE = 1.613848
## q = 6 AIC = 4637.489 BIC = 4677.837 MASE = 1.607532
## q = 7 AIC = 4632.716 BIC = 4677.532 MASE = 1.607042
## q = 8 AIC = 4625.986 BIC = 4675.267 MASE = 1.604806
## q = 9 AIC = 4615.084 BIC = 4668.827 MASE = 1.593121
## q = 10 AIC = 4602.658 BIC = 4660.858 MASE = 1.577996
## q = 11 AIC = 4590.961 BIC = 4653.617 MASE = 1.562127
## q = 12 AIC = 4578.787 BIC = 4645.895 MASE = 1.5516
```

- The model provides a poor overall fit with an adjusted R-squared of only 0.3077. While the model is statistically significant and free from severe multicollinearity (VIF < 10 for all terms), only the current and 12-month lagged Precipitation coefficients are significant. Crucially, the model is invalid for forecasting because its residuals exhibit strong, unmodeled autocorrelation at the non-seasonal (Lag 1) and seasonal (Lag 12) periods, indicating that the internal time-dependencies of the Solar Radiation series have not been captured.
- Therefore, the next model will be examined to find the best fit.

```r
model1<-dlm(x = as.vector(solar_data$precipitation),
                  y = as.vector(solar_data$solar), q = 12) #apply model
summarise(model1)
```

```
##
## Call:
## lm(formula = model.formula, data = design)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -18.563  -5.239  -0.796   4.137  32.430
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  19.5164     1.1151  17.501  < 2e-16 ***
## x.t          -5.8876     1.9508  -3.018  0.00265 **
## x.1           0.9993     2.5647   0.390  0.69694
## x.2           0.4343     2.5571   0.170  0.86520
## x.3           1.8763     2.5580   0.734  0.46352
## x.4           1.7459     2.5587   0.682  0.49529
## x.5           3.3279     2.5601   1.300  0.19410
## x.6           0.7751     2.5617   0.303  0.76230
## x.7           1.7937     2.5615   0.700  0.48402
## x.8           0.2827     2.5593   0.110  0.91207
## x.9          -1.1022     2.5615  -0.430  0.66712
## x.10         -1.9333     2.5508  -0.758  0.44880
## x.11         -0.5613     2.5532  -0.220  0.82605
## x.12         -5.3492     1.9216  -2.784  0.00553 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.181 on 634 degrees of freedom
## Multiple R-squared:  0.3216, Adjusted R-squared:  0.3077
## F-statistic: 23.12 on 13 and 634 DF,  p-value: < 2.2e-16
##
## AIC and BIC values for the model:
##        AIC      BIC
## 1 4578.787 4645.895
```




```
##   x.t   x.1   x.2   x.3   x.4   x.5   x.6   x.7   x.8   x.9  x.10  x.11
x.12
## FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
FALSE
```

b. Polynomial DLM

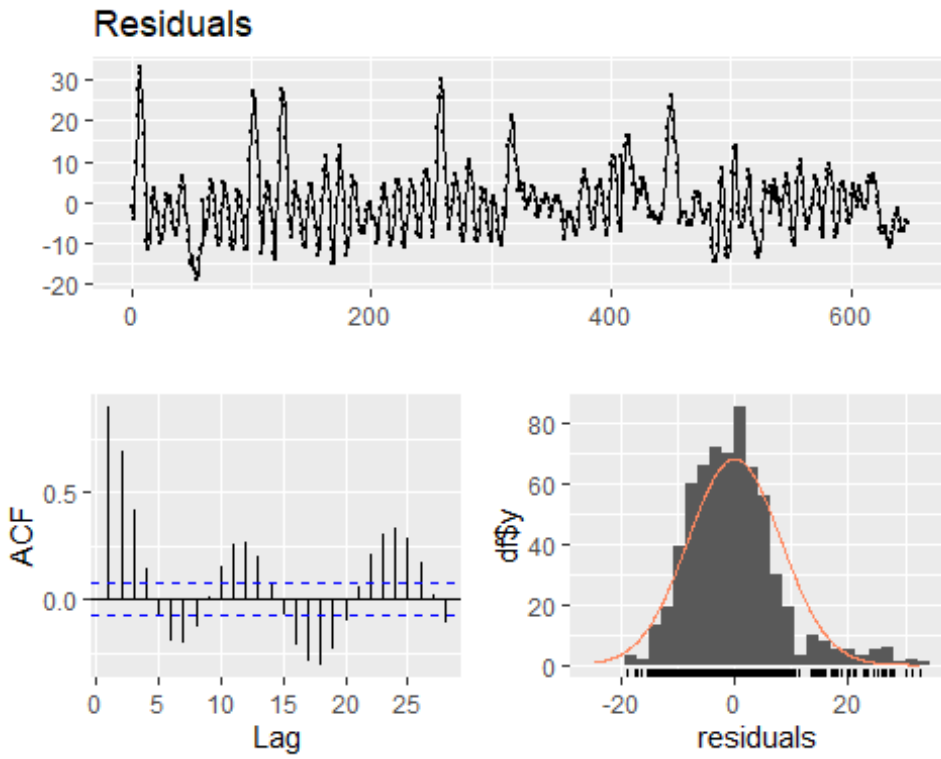
- The Polynomial DLM was tested to improve model parsimony. By assuming the lag coefficients follow a quadratic polynomial (k=2), both Model 2 (q=12) and Model 2.1 (q=11) achieved nearly the same level of explainability (Adjusted R-squared approx. 0.31) as the full DLM but used far fewer parameters, confirming the efficiency of the polynomial assumption. Model 2 (q=12) was marginally the best of the two. However, the PDLM structure is still fundamentally flawed because it fails the residual checks: the errors remain highly autocorrelated at Lags 1 and 12. Since the model errors are not white noise, the PDLMs are not valid for generating reliable forecasts. The next step is required to address this residual seasonality.

```
model2 <- polyDlm(x = as.vector(solar_data$precipitation),
                  y = as.vector(solar_data$solar), q=12, k=2, show.beta =
TRUE)

## Estimates and t-tests for beta coefficients:
##         Estimate Std. Error t value  P(>|t|)
## beta.0    -1.820      0.393   -4.63 4.41e-06
## beta.1    -0.402      0.311   -1.29 1.97e-01
## beta.2     0.702      0.261    2.69 7.42e-03
```

```
## beta.3     1.490      0.240    6.22 9.28e-10
## beta.4     1.970      0.237    8.31 5.71e-16
## beta.5     2.130      0.239    8.89 6.28e-18
## beta.6     1.970      0.240    8.22 1.18e-15
## beta.7     1.510      0.237    6.36 3.90e-10
## beta.8     0.726      0.232    3.13 1.84e-03
## beta.9    -0.370      0.233   -1.58 1.14e-01
## beta.10   -1.780      0.254   -7.01 5.94e-12
## beta.11   -3.500      0.304  -11.50 4.37e-28
## beta.12   -5.540      0.386  -14.30 1.31e-40
```

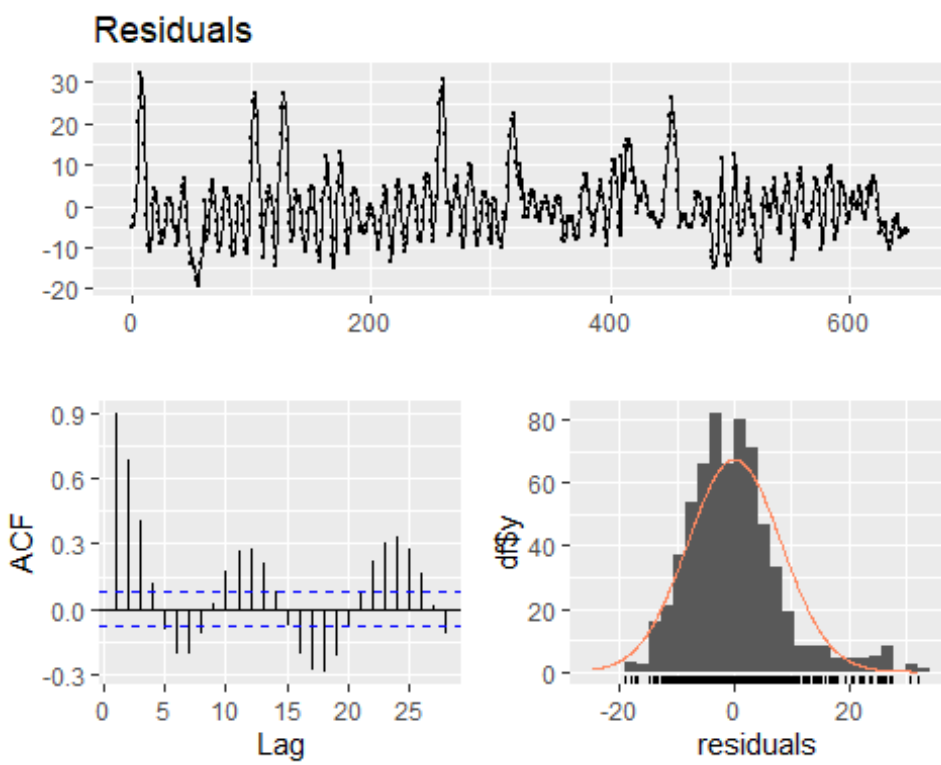
```r
summarise(model2)
```

```
##
## Call:
## "Y ~ (Intercept) + X.t"
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -18.689  -5.452  -0.686   4.129  32.797
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 19.22679    1.10784  17.355  < 2e-16 ***
## z.t0        -1.81945    0.39287  -4.631  4.4e-06 ***
## z.t1         1.57478    0.13106  12.015  < 2e-16 ***
## z.t2        -0.15708    0.01019 -15.409  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.175 on 644 degrees of freedom
## Multiple R-squared:  0.3119, Adjusted R-squared:  0.3087
## F-statistic: 97.31 on 3 and 644 DF,  p-value: < 2.2e-16
```




```
##  z.t0  z.t1  z.t2
## FALSE  TRUE  TRUE

model2.1 <- polyDlm(x = as.vector(solar_data$precipitation),
                    y = as.vector(solar_data$solar), q=11, k=2, show.beta =
TRUE)

## Estimates and t-tests for beta coefficients:
##          Estimate Std. Error t value  P(>|t|)
## beta.0    -3.5800      0.400  -8.950 3.97e-18
## beta.1    -1.5200      0.305  -4.990 7.76e-07
## beta.2     0.0986      0.252   0.391 6.96e-01
## beta.3     1.2900      0.235   5.470 6.54e-08
## beta.4     2.0400      0.237   8.590 6.85e-17
## beta.5     2.3600      0.242   9.720 6.23e-21
## beta.6     2.2400      0.242   9.270 2.86e-19
## beta.7     1.6900      0.235   7.180 1.99e-12
## beta.8     0.7050      0.232   3.040 2.46e-03
## beta.9    -0.7140      0.248  -2.880 4.13e-03
## beta.10   -2.5700      0.301  -8.530 1.06e-16
## beta.11   -4.8500      0.396 -12.300 3.82e-31

summarise(model2.1)

##
## Call:
## "Y ~ (Intercept) + X.t"
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.160  -5.251  -0.685   4.004  31.965
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 19.13614    1.09602  17.460   <2e-16 ***
## z.t0        -3.57832    0.39999  -8.946   <2e-16 ***
## z.t1         2.27277    0.14969  15.183   <2e-16 ***
## z.t2        -0.21717    0.01292 -16.813   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.189 on 645 degrees of freedom
## Multiple R-squared:  0.3105, Adjusted R-squared:  0.3073
## F-statistic: 96.81 on 3 and 645 DF,  p-value: < 2.2e-16
```



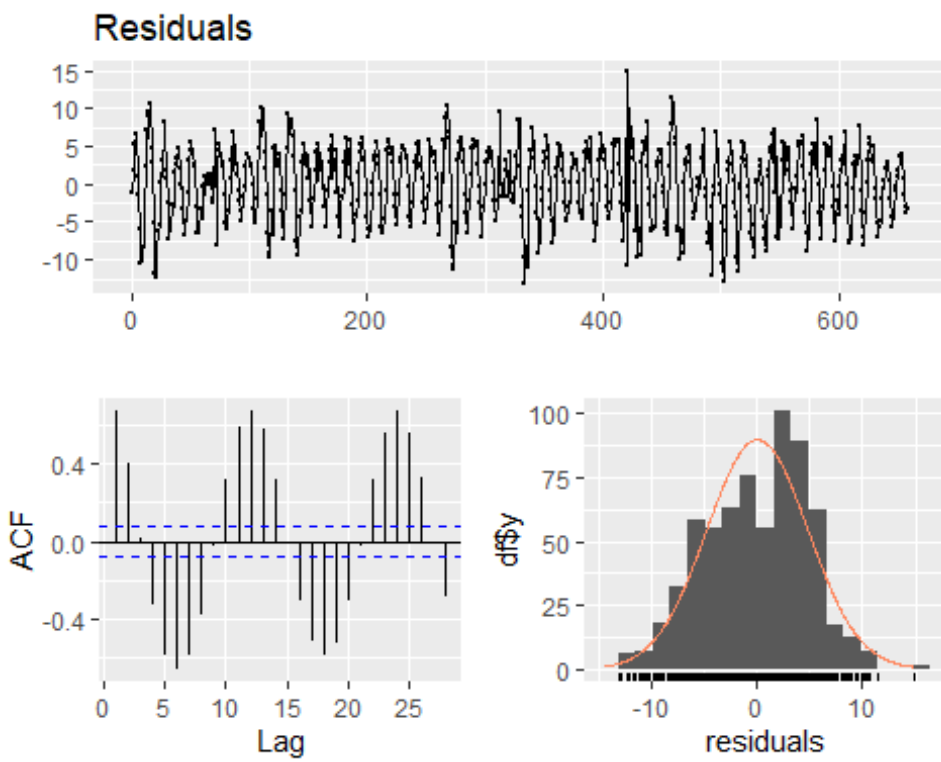

```
##  z.t0  z.t1  z.t2
## FALSE  TRUE  TRUE
```

c. *Koyck transformation*

- The Koyck DLM represents a significant advancement over the previous models, achieving a high Adjusted R-squared of 0.7591 while dramatically reducing the residual standard error to 4.814. This is due to the inherent structure of the Koyck model, which uses the lagged dependent variable to efficiently capture the long-

term effects and high persistence within the solar radiation series. Both the concurrent Precipitation and the lagged Solar Radiation coefficients are highly significant. However, the model is still invalid for forecasting because the Ljung-Box test strongly rejects the white noise assumption, confirmed by the remaining autocorrelation spikes in the residual ACF plot, particularly the unmodeled seasonal component at Lag 12.

```
model3 <- koyckDlm(x = as.vector(solar_data$precipitation),
                   y = as.vector(solar_data$solar))
summarise(model3)

##
## Call:
## "Y ~ (Intercept) + Y.1 + X.t"
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.0926  -3.5961   0.3176   3.6103  14.8399
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.23925    0.76549  -2.925  0.00356 **
## Y.1          0.98546    0.02424  40.650  < 2e-16 ***
## X.t          5.34684    0.84383   6.336 4.37e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.814 on 656 degrees of freedom
## Multiple R-Squared: 0.7598,  Adjusted R-squared: 0.7591
## Wald test:  1104 on 2 and 656 DF,  p-value: < 2.2e-16
##
## Diagnostic tests:
## NULL
##
##                              alpha     beta       phi
## Geometric coefficients:  -154.0203 5.346844 0.9854613
```



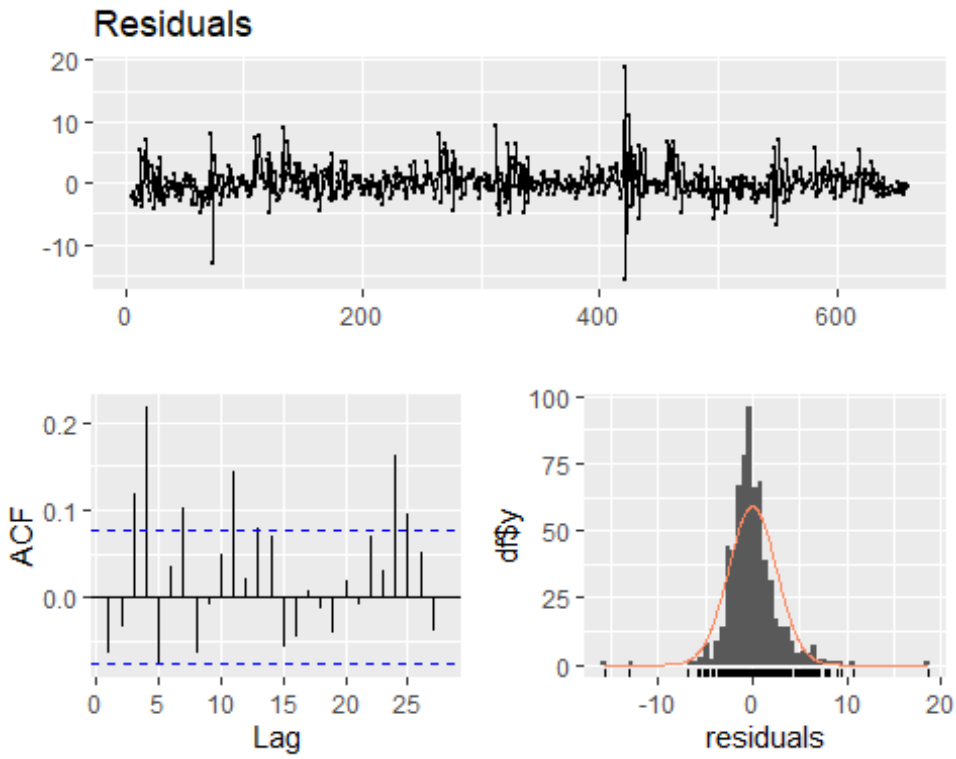

```
##
##  Ljung-Box test
##
## data:  Residuals
## Q* = 1413.2, df = 10, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 10

##   Y.1   X.t
## FALSE FALSE
```

d. Autoregressive DLM

- A loop was run to see which autoregressive DLM model has the lowest AIC, BIC, and MASE. Comparing among models, there are three models selected including ARDL(2,5), ARDL(4,5), ARDL(5,5)
- The Autoregressive Distributed Lag (ARDL) model represents the definitive refinement of the time series regression approach. By systematically testing the model orders, ARDL(4,5) and ARDL(5,5) demonstrated superior performance. Both models achieved an outstanding Adjusted R-squared of over 0.93, meaning more than 93% of the variance in Solar Radiation is now explained. Most importantly, the ARDL structure successfully addressed all previous residual issues, resulting in white noise errors, validating the model for forecasting. The ARDL(5,5) yielded the lowest MASE score of 0.44793, making it the best time series regression model tested. The model shows the Solar Radiation series' own historical values (Yt-1, Yt-

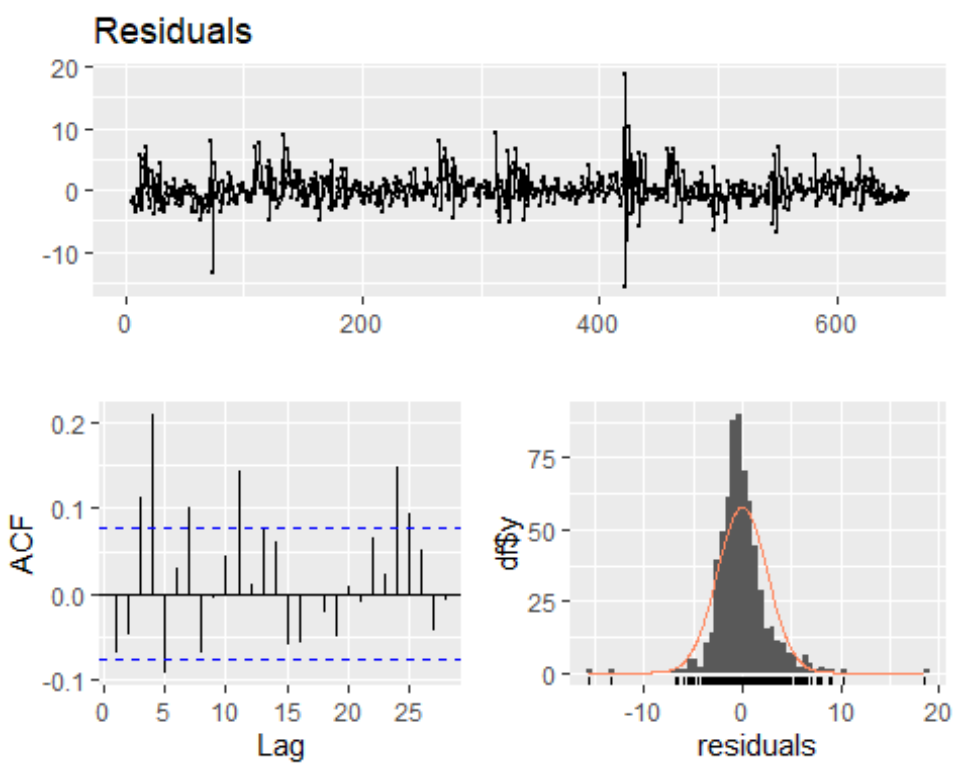
3, Yt-4, Yt-5) are the primary drivers, with Precipitation having a weak, delayed influence at a four-month lag (Xt-4).

```
for (i in 1:5){ # loop to identify the lag with the best AIC, BIC and MASE
  for(j in 5){
    model4.1 = ardlDlm(x = as.vector(solar_data$precipitation),
                       y = as.vector(solar_data$solar), p = i , q = j)
    cat("p =", i, "q =", j, "AIC =", AIC(model4.1$model), "BIC =",
BIC(model4.1$model), "MASE =", MASE(model4.1)$MASE, "\n")
  }
}

## p = 1 q = 5 AIC = 3100.283 BIC = 3140.644 MASE = 0.450425
## p = 2 q = 5 AIC = 3097.288 BIC = 3142.134 MASE = 0.4503599
## p = 3 q = 5 AIC = 3098.808 BIC = 3148.139 MASE = 0.4502885
## p = 4 q = 5 AIC = 3096.024 BIC = 3149.839 MASE = 0.4479481
## p = 5 q = 5 AIC = 3097.877 BIC = 3156.177 MASE = 0.4479311

model4.25 <-ardlDlm(x = as.vector(solar_data$precipitation), y =
as.vector(solar_data$solar), p=2, q=5) #apply model
summarise(model4.25)

##
## Time series regression with "ts" data:
## Start = 6, End = 660
##
## Call:
## dynlm(formula = as.formula(model.text), data = data, start = 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.6548  -1.4360  -0.2439   1.0820  18.7279
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.13009    0.40777   5.224 2.37e-07 ***
## X.t         -0.56938    0.54642  -1.042 0.297791
## X.1          0.99538    0.76194   1.306 0.191894
## X.2          1.23236    0.55418   2.224 0.026508 *
## Y.1          1.27096    0.03840  33.094  < 2e-16 ***
## Y.2         -0.02873    0.06247  -0.460 0.645713
## Y.3         -0.40207    0.06030  -6.668 5.58e-11 ***
## Y.4         -0.22362    0.06219  -3.596 0.000348 ***
## Y.5          0.22001    0.03773   5.831 8.72e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.552 on 646 degrees of freedom
## Multiple R-squared:  0.9333, Adjusted R-squared:  0.9324
## F-statistic:  1129 on 8 and 646 DF,  p-value: < 2.2e-16
```




```
##       X.t L(X.t, 1) L(X.t, 2) L(y.t, 1) L(y.t, 2) L(y.t, 3) L(y.t, 4)
L(y.t, 5)
##     FALSE     FALSE     FALSE      TRUE      TRUE      TRUE      TRUE
TRUE

model4.45 <-ardlDlm(x = as.vector(solar_data$precipitation),y =
as.vector(solar_data$solar), p=4, q=5)
summarise(model4.45)

##
## Time series regression with "ts" data:
## Start = 6, End = 660
##
## Call:
## dynlm(formula = as.formula(model.text), data = data, start = 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5901  -1.3826  -0.2867   1.0526  18.5709
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.46103    0.43729   5.628 2.72e-08 ***
## X.t         -0.61439    0.54768  -1.122 0.262364
## X.1          0.78469    0.77617   1.011 0.312409
## X.2          1.28470    0.79026   1.626 0.104509
## X.3          0.80457    0.77946   1.032 0.302355
```

```
## X.4         -1.20750    0.55570  -2.173 0.030148 *
## Y.1          1.27195    0.03849  33.049  < 2e-16 ***
## Y.2         -0.01868    0.06249  -0.299 0.765112
## Y.3         -0.40480    0.06019  -6.725 3.87e-11 ***
## Y.4         -0.23201    0.06221  -3.729 0.000209 ***
## Y.5          0.21746    0.03772   5.766 1.26e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.546 on 644 degrees of freedom
## Multiple R-squared:  0.9338, Adjusted R-squared:  0.9328
## F-statistic: 908.5 on 10 and 644 DF,  p-value: < 2.2e-16
```

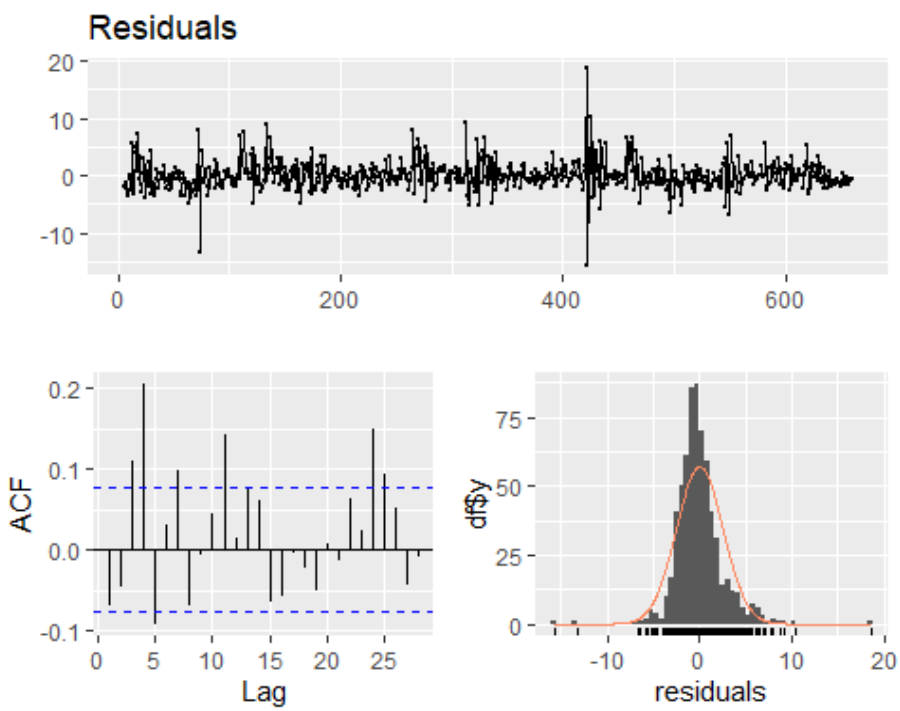



```
##       X.t L(X.t, 1) L(X.t, 2) L(X.t, 3) L(X.t, 4) L(y.t, 1) L(y.t, 2)
L(y.t, 3)
##     FALSE     FALSE     FALSE     FALSE     FALSE      TRUE      TRUE
TRUE
## L(y.t, 4) L(y.t, 5)
##      TRUE      TRUE
```

```
model4.55 <-ardlDlm(x = as.vector(solar_data$precipitation),y =
as.vector(solar_data$solar), p=5, q=5)
summarise(model4.55)
```

```
##
## Time series regression with "ts" data:
## Start = 6, End = 660
```

```
##
## Call:
## dynlm(formula = as.formula(model.text), data = data, start = 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5959  -1.3825  -0.2646   1.0410  18.5812
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.50740    0.45434   5.519 4.96e-08 ***
## X.t         -0.61416    0.54804  -1.121 0.262863
## X.1          0.78299    0.77670   1.008 0.313788
## X.2          1.26543    0.79241   1.597 0.110772
## X.3          0.75184    0.79227   0.949 0.342998
## X.4         -1.00181    0.77678  -1.290 0.197617
## X.5         -0.21024    0.55439  -0.379 0.704639
## Y.1          1.27063    0.03867  32.861  < 2e-16 ***
## Y.2         -0.01727    0.06264  -0.276 0.782907
## Y.3         -0.40297    0.06043  -6.669 5.56e-11 ***
## Y.4         -0.23273    0.06229  -3.737 0.000203 ***
## Y.5          0.21571    0.03802   5.673 2.12e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.548 on 643 degrees of freedom
## Multiple R-squared:  0.9338, Adjusted R-squared:  0.9327
## F-statistic: 824.9 on 11 and 643 DF,  p-value: < 2.2e-16
```



```
##       X.t L(X.t, 1) L(X.t, 2) L(X.t, 3) L(X.t, 4) L(X.t, 5) L(y.t, 1)
L(y.t, 2)
##     FALSE     FALSE     FALSE     FALSE     FALSE     FALSE      TRUE
TRUE
## L(y.t, 3) L(y.t, 4) L(y.t, 5)
##      TRUE      TRUE      TRUE
```

- The evaluation of all Distributed Lag Model (DLM) variants confirmed that only the Autoregressive Distributed Lag (ARDL) models provided a valid, accurate fit with white noise residuals. The simpler Finite, Polynomial, and Koyck DLMs were rejected due to their inability to model the strong autocorrelation and seasonality present in the Solar Radiation series, resulting in poor MASE scores (ranging from 1.03 to 1.57). The ARDL (p=5, q=5) model delivered the best forecast performance among all regression methods, achieving the lowest Mean Absolute Scaled Error (MASE = 0.4479311). This model is now the leading candidate from the time series regression group.

Comparative board

```r
# List of models
attr(model3$model,"class") = "lm"
model_results <- data.frame( #compartive board
  Model = c("Finite DLM", "Polynomial DLM (q=11, k=2)", "Polynomial DLM
(q=12, k=2)", "Koyck", "ADLM (p=2, q=5)", "ADLM (p=4, q=5)", "ADLM (p=5,
q=5)"),
  AIC = c(AIC(model1$model), AIC(model2.1$model), AIC(model2$model),
AIC(model3$model), AIC(model4.25$model), AIC(model4.45$model),
AIC(model4.55$model)),
  BIC = c(BIC(model1$model), BIC(model2.1$model), BIC(model2$model),
BIC(model3$model), BIC(model4.25$model), BIC(model4.45$model),
BIC(model4.55$model)),
  MASE = c(MASE(model1)$MASE, MASE(model2.1)$MASE, MASE(model2)$MASE,
MASE(model3)$MASE, MASE(model4.25)$MASE, MASE(model4.45)$MASE,
MASE(model4.55)$MASE)
)
reg_results <- model_results[order(model_results$MASE), ]
# Display the table
print(reg_results)
```

```
##                         Model      AIC      BIC      MASE
## 7            ADLM (p=5, q=5) 3097.877 3156.177 0.4479311
## 6            ADLM (p=4, q=5) 3096.024 3149.839 0.4479481
## 5            ADLM (p=2, q=5) 3097.288 3142.134 0.4503599
## 4                      Koyck 3946.476 3964.439 1.0324829
## 1                 Finite DLM 4578.787 4645.895 1.5516004
## 3 Polynomial DLM (q=12, k=2) 4567.969 4590.339 1.5630351
## 2 Polynomial DLM (q=11, k=2) 4577.156 4599.533 1.5704569
```

2. Dynamic Models

- The two Dynamic Linear Models tested were based on the high-performing ARDL(5,5) structure, enhanced by explicit seasonal dummy variables.
    - Model 1: ARDL + Seasonality
    - Model 2: ARDL + Trend + Seasonality

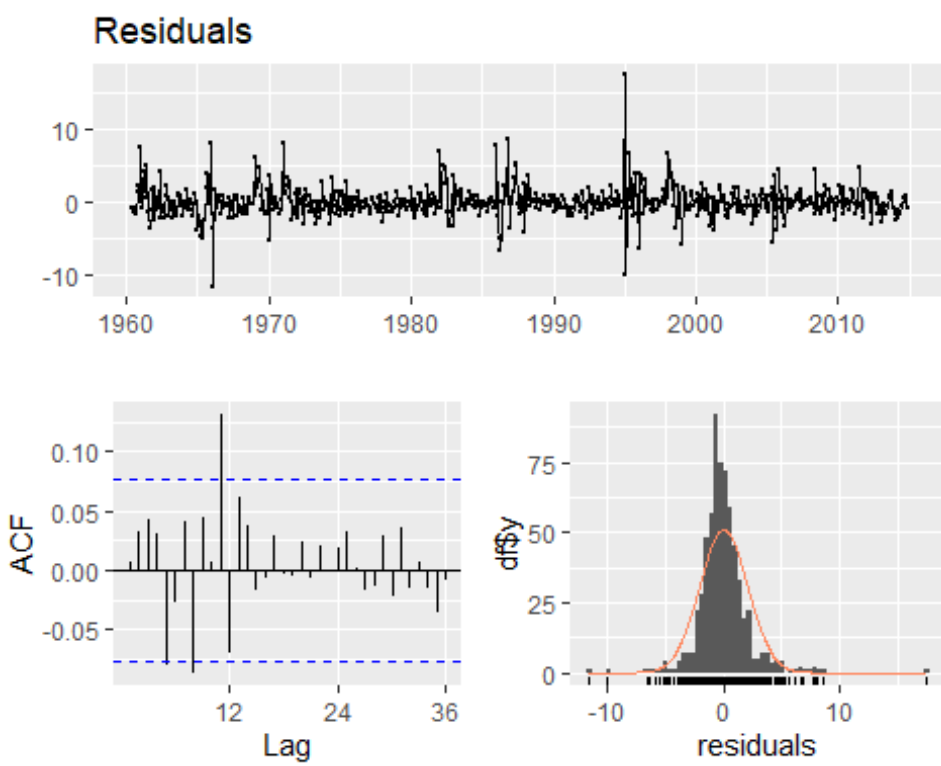
```
Y.t = solar_ts
X.t = precip_ts
dyn1 <- dynlm(Y.t ~ L(Y.t, k=1:5) +
                            L(X.t, k=0:5) +
                            season(Y.t))

dyn2 <- dynlm(Y.t ~ L(Y.t, k=1:5) +
                            L(X.t, k=0:5) +
                            trend(Y.t) +
                            season(Y.t))
```

- This model is highly successful at fitting the data (Adjusted R-squared> 0.95) by correctly identifying the importance of the series' internal memory and the strong seasonal cycle. However, its major weakness is that it fails the Breusch-Godfrey test for white noise residuals, meaning it cannot be reliably used for forecasting. The Precipitation offers no unique, significant explanatory power.

```
summary(dyn1)

##
## Time series regression with "ts" data:
## Start = 1960(6), End = 2014(12)
##
## Call:
## dynlm(formula = Y.t ~ L(Y.t, k = 1:5) + L(X.t, k = 0:5) + season(Y.t))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.6291  -0.9456  -0.1343   0.8182  17.4776
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        5.35136    0.57612   9.289  < 2e-16 ***
## L(Y.t, k = 1:5)1   0.99799    0.03957  25.220  < 2e-16 ***
## L(Y.t, k = 1:5)2   0.13283    0.05585   2.378   0.0177 *
## L(Y.t, k = 1:5)3  -0.07308    0.05596  -1.306   0.1920
## L(Y.t, k = 1:5)4  -0.12173    0.05578  -2.183   0.0294 *
## L(Y.t, k = 1:5)5  -0.07351    0.03961  -1.856   0.0639 .
## L(X.t, k = 0:5)0  -0.19627    0.50314  -0.390   0.6966
## L(X.t, k = 0:5)1  -0.79507    0.69438  -1.145   0.2526
## L(X.t, k = 0:5)2   0.44683    0.69978   0.639   0.5234
## L(X.t, k = 0:5)3   0.13026    0.69779   0.187   0.8520
```

```
## L(X.t, k = 0:5)4  0.35501    0.69107   0.514   0.6076
## L(X.t, k = 0:5)5 -0.79286    0.49901  -1.589   0.1126
## season(Y.t)Feb    0.50441    0.45818   1.101   0.2714
## season(Y.t)Mar    2.10013    0.51597   4.070 5.29e-05 ***
## season(Y.t)Apr   -0.22104    0.60389  -0.366   0.7145
## season(Y.t)May    0.42245    0.65705   0.643   0.5205
## season(Y.t)Jun   -1.57494    0.69802  -2.256   0.0244 *
## season(Y.t)Jul   -3.25257    0.72081  -4.512 7.64e-06 ***
## season(Y.t)Aug   -6.26589    0.70658  -8.868  < 2e-16 ***
## season(Y.t)Sep   -7.52509    0.69623 -10.808  < 2e-16 ***
## season(Y.t)Oct   -6.98149    0.63866 -10.932  < 2e-16 ***
## season(Y.t)Nov   -4.95272    0.54590  -9.073  < 2e-16 ***
## season(Y.t)Dec   -2.29478    0.45741  -5.017 6.83e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.111 on 632 degrees of freedom
## Multiple R-squared:  0.9554, Adjusted R-squared:  0.9538
## F-statistic: 614.8 on 22 and 632 DF,  p-value: < 2.2e-16
```

```
checkresiduals(dyn1)
```

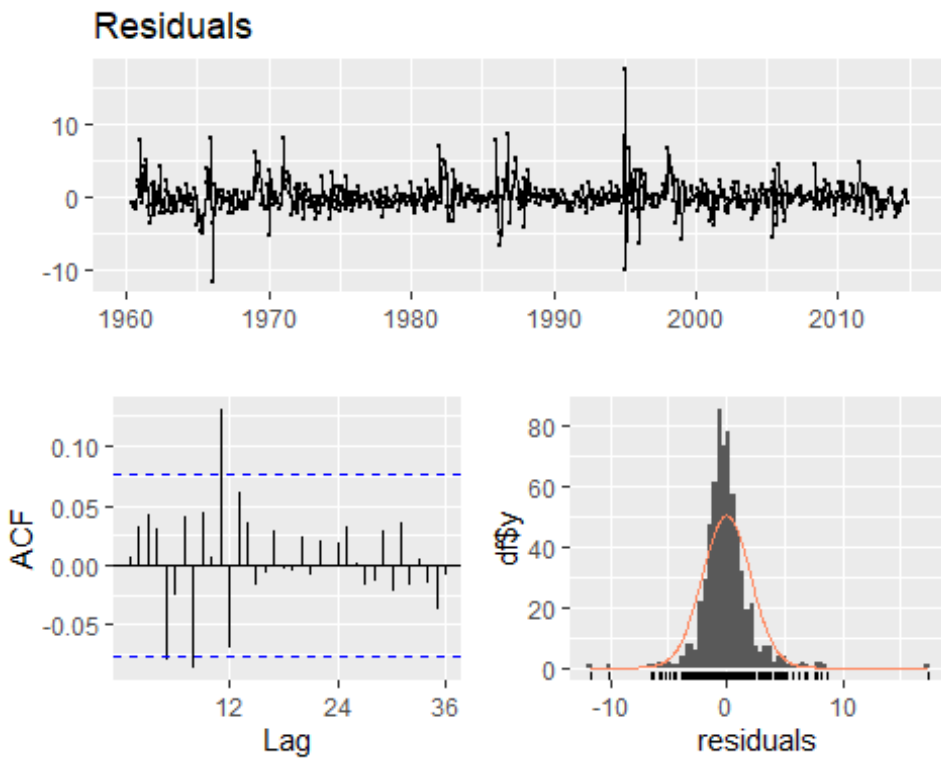



```
##
##  Breusch-Godfrey test for serial correlation of order up to 26
##
## data:  Residuals
## LM test = 44.395, df = 26, p-value = 0.0137
```

- This model performed no better than dyn1 in terms of fit or residual validity, as confirmed by the nearly identical Adjusted R-squared and the failed Breusch-Godfrey test. Since the added trend term was highly insignificant, Model 2 is unnecessarily complex and less parsimonious than Model 1.

```
summary(dyn2)

##
## Time series regression with "ts" data:
## Start = 1960(6), End = 2014(12)
##
## Call:
## dynlm(formula = Y.t ~ L(Y.t, k = 1:5) + L(X.t, k = 0:5) + trend(Y.t) +
##     season(Y.t))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.6047  -0.9403  -0.1408   0.8093  17.4692
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       5.324206   0.588436   9.048  < 2e-16 ***
## L(Y.t, k = 1:5)1  0.997938   0.039601  25.200  < 2e-16 ***
## L(Y.t, k = 1:5)2  0.132899   0.055894   2.378   0.0177 *
## L(Y.t, k = 1:5)3 -0.073029   0.055998  -1.304   0.1927
## L(Y.t, k = 1:5)4 -0.121723   0.055817  -2.181   0.0296 *
## L(Y.t, k = 1:5)5 -0.073665   0.039642  -1.858   0.0636 .
## L(X.t, k = 0:5)0 -0.199924   0.503769  -0.397   0.6916
## L(X.t, k = 0:5)1 -0.794937   0.694899  -1.144   0.2531
## L(X.t, k = 0:5)2  0.445811   0.700321   0.637   0.5246
## L(X.t, k = 0:5)3  0.130079   0.698318   0.186   0.8523
## L(X.t, k = 0:5)4  0.354029   0.691605   0.512   0.6089
## L(X.t, k = 0:5)5 -0.795253   0.499493  -1.592   0.1119
## trend(Y.t)        0.001212   0.005254   0.231   0.8176
## season(Y.t)Feb    0.504368   0.458522   1.100   0.2718
## season(Y.t)Mar    2.100028   0.516353   4.067 5.36e-05 ***
## season(Y.t)Apr   -0.221462   0.604350  -0.366   0.7142
## season(Y.t)May    0.421068   0.657568   0.640   0.5222
## season(Y.t)Jun   -1.577129   0.698603  -2.258   0.0243 *
## season(Y.t)Jul   -3.255741   0.721478  -4.513 7.64e-06 ***
## season(Y.t)Aug   -6.269193   0.707250  -8.864  < 2e-16 ***
## season(Y.t)Sep   -7.528334   0.696896 -10.803  < 2e-16 ***
## season(Y.t)Oct   -6.983684   0.639205 -10.926  < 2e-16 ***
## season(Y.t)Nov   -4.954294   0.546349  -9.068  < 2e-16 ***
## season(Y.t)Dec   -2.295349   0.457761  -5.014 6.93e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.112 on 631 degrees of freedom
```

```
## Multiple R-squared:  0.9554, Adjusted R-squared:  0.9537
## F-statistic: 587.2 on 23 and 631 DF,  p-value: < 2.2e-16

checkresiduals(dyn2)
```




```
##
##  Breusch-Godfrey test for serial correlation of order up to 27
##
## data:  Residuals
## LM test = 44.689, df = 27, p-value = 0.01753
```

- The comprehensive evaluation of all time series regression models leads to the selection of the ARDL (p=5, q=5) model as the best overall candidate from this methodology group. While the Dynamic Linear Model 1 (DLM 1) achieved a superior model fit, evidenced by the lowest AIC (2861.98) and highest Adjusted R-squared (0.9538), both DLM models were statistically invalid due to a failed Breusch-Godfrey test, meaning significant autocorrelation remains in the residuals.
- In contrast, the ARDL (p=5, q=5) model achieved the lowest Mean Absolute Scaled Error (MASE = 0.44793), demonstrating the best forecast accuracy among all valid models. Crucially, its residuals were confirmed to be white noise, making it a reliable and valid model for inference.
- The final best model from the regression analysis is the ADLM (p=5, q=5)

```
dnm_results <- data.frame( #comparative board for Dynamic Models
  Model = c("Dynamic Linear 1", "Dynamic Linear 2"),
```

```
  AIC = c(AIC(dyn1), AIC(dyn2)),
  BIC = c(BIC(dyn1), BIC(dyn2)),
  MASE = c(NA, NA))
compared_board <- rbind(reg_results, dnm_results) #combine regression models
+ dynamic models
compared_board <- compared_board[order(compared_board$MASE), ]
compared_board

##                          Model      AIC      BIC      MASE
## 7             ADLM (p=5, q=5) 3097.877 3156.177 0.4479311
## 6             ADLM (p=4, q=5) 3096.024 3149.839 0.4479481
## 5             ADLM (p=2, q=5) 3097.288 3142.134 0.4503599
## 4                       Koyck 3946.476 3964.439 1.0324829
## 1                  Finite DLM 4578.787 4645.895 1.5516004
## 3  Polynomial DLM (q=12, k=2) 4567.969 4590.339 1.5630351
## 2  Polynomial DLM (q=11, k=2) 4577.156 4599.533 1.5704569
## 11           Dynamic Linear 1 2861.980 2969.611        NA
## 21           Dynamic Linear 2 2863.925 2976.041        NA
```

3. Exponential smoothing

- The Holt-Winters models revealed a dramatic improvement in forecast accuracy compared to all preceding Time Series Regression methods. All four HW models achieved a MASE score significantly lower than the best ARDL model
    - The solar radiation series is highly stable and seasonal, making the Exponential Smoothing approach far superior to regression-based models
    - The Multiplicative seasonal variants were superior to the Additive variants, confirming that the magnitude of the seasonal cycle grows proportionally with the level of the series
    - The two best models are the HW Multiplicative Damped and the HW Multiplicative, which will be looked into

```
fit1.add <- hw(solar_ts, seasonal = "additive")
fit2.add.damp <- hw(solar_ts, seasonal = "additive", damped = TRUE)
fit3.mul <- hw(solar_ts, seasonal = "multiplicative")
fit4.mul.damp <- hw(solar_ts, seasonal = "multiplicative", damped = TRUE)

hw_results <- data.frame( #comparative board for exponential smoothing
  Model = c("HW Additive", "HW Additive Damped", "HW Multiplicative", "HW
Multiplicative Damped"),
  AIC = c(fit1.add$model$aic, fit2.add.damp$model$aic, fit3.mul$model$aic,
fit4.mul.damp$model$aic),
  BIC = c(fit1.add$model$bic, fit2.add.damp$model$bic, fit3.mul$model$bic,
fit4.mul.damp$model$bic),
  MASE = c(accuracy(fit1.add)[,"MASE"][1],
accuracy(fit2.add.damp)[,"MASE"][1],
           accuracy(fit3.mul)[,"MASE"][1],
accuracy(fit4.mul.damp)[,"MASE"][1]))
```

```r
compared_board <- rbind(compared_board, hw_results)
compared_board <- compared_board[order(compared_board$MASE), ]
compared_board
```

Comparative board

```
##                          Model      AIC      BIC      MASE
## 41   HW Multiplicative Damped 6366.701 6447.561 0.2035619
## 31          HW Multiplicative 6648.746 6725.114 0.2233077
## 22         HW Additive Damped 5428.422 5509.282 0.2461797
## 12                HW Additive 5434.708 5511.076 0.2471600
## 7             ADLM (p=5, q=5) 3097.877 3156.177 0.4479311
## 6             ADLM (p=4, q=5) 3096.024 3149.839 0.4479481
## 5             ADLM (p=2, q=5) 3097.288 3142.134 0.4503599
## 4                       Koyck 3946.476 3964.439 1.0324829
## 1                  Finite DLM 4578.787 4645.895 1.5516004
## 3  Polynomial DLM (q=12, k=2) 4567.969 4590.339 1.5630351
## 2  Polynomial DLM (q=11, k=2) 4577.156 4599.533 1.5704569
## 11           Dynamic Linear 1 2861.980 2969.611        NA
## 21           Dynamic Linear 2 2863.925 2976.041        NA
```

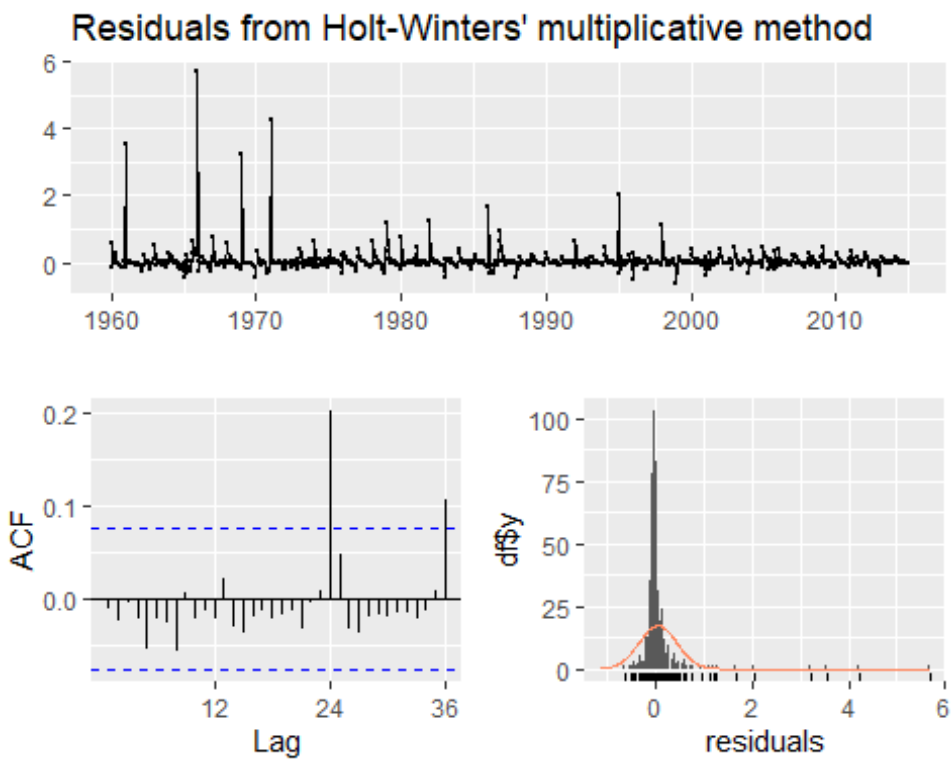
- This model is an extremely accurate descriptive model (MASE 0.223) that correctly identifies multiplicative seasonality and a negligible trend.
- However, it is statistically invalid for forecasting because it fails the Ljung-Box test. The remaining autocorrelation indicates that the model is mis-specified, and its prediction intervals will be unreliable

```r
summary(fit3.mul)
```

```
##
## Forecast method: Holt-Winters' multiplicative method
##
## Model Information:
## Holt-Winters' multiplicative method
##
## Call:
## hw(y = solar_ts, seasonal = "multiplicative")
##
##   Smoothing parameters:
##     alpha = 0.8251
##     beta  = 1e-04
##     gamma = 0.0259
##
##   Initial states:
##     l = 10.5659
##     b = -0.2371
##     s = 0.5024 0.6292 0.9319 1.2229 1.4923 1.6309
##            1.5606 1.3672 1.0278 0.8128 0.5106 0.3114
##
```

```
##   sigma:  0.3971
##
##      AIC     AICc      BIC
## 6648.746 6649.699 6725.114
##
## Error measures:
##                      ME    RMSE      MAE        MPE     MAPE      MASE
## Training set 0.04281136 2.12539 1.359297 -0.4896895 10.87401 0.2233077
##                    ACF1
## Training set 0.06551473
##
## Forecasts:
##          Point Forecast        Lo 80     Hi 80       Lo 95     Hi 95
## Jan 2015       5.610335   2.75534413  8.465326   1.2440033  9.976666
## Feb 2015       6.512806   2.03809262 10.987520  -0.3306776 13.356290
## Mar 2015       8.786120   1.33667747 16.235562  -2.6068190 20.179058
## Apr 2015      10.192785  -0.02672876 20.412298  -5.4366123 25.822181
## May 2015      12.512597  -1.96157245 26.986766  -9.6237347 34.648928
## Jun 2015      14.061609  -4.41048190 32.533701 -14.1890164 42.312235
## Jul 2015      14.502088  -6.89909775 35.903274 -18.2282011 47.232377
## Aug 2015      12.945620  -8.35024040 34.241481 -19.6235881 45.514829
## Sep 2015      10.352210  -8.52365174 29.228072 -18.5159294 39.220350
## Oct 2015       7.418279  -7.51119146 22.347750 -15.4143760 30.250935
## Nov 2015       4.989730  -6.05900058 16.038460 -11.9078451 21.887305
## Dec 2015       3.871776  -5.53876625 13.282319 -10.5204066 18.263960
## Jan 2016       4.199400  -7.01753622 15.416336 -12.9554236 21.354224
## Feb 2016       4.838892  -9.30305408 18.980837 -16.7893479 26.467131
## Mar 2016       6.477174 -14.21845674 27.172805 -25.1740619 38.128410
## Apr 2016       7.452633 -18.56883703 33.474103 -32.3437711 47.249037
## May 2016       9.069749 -25.52978297 43.669280 -43.8456686 61.985166
## Jun 2016      10.099485 -31.99838423 52.197355 -54.2836502 74.482621
## Jul 2016      10.315199 -36.68017471 57.310572 -61.5580226 82.188420
## Aug 2016       9.113765 -36.29212001 54.519651 -60.3285438 78.556075
## Sep 2016       7.208700 -32.09306283 46.510463 -52.8981594 67.315559
## Oct 2016       5.105869 -25.38360449 35.595343 -41.5237568 51.735495
## Nov 2016       3.391945 -18.81647619 25.600367 -30.5729044 37.356795
## Dec 2016       2.597256 -16.07148862 21.266001 -25.9541251 31.148637
```

```
checkresiduals(fit3.mul)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Holt-Winters' multiplicative method
## Q* = 38.585, df = 24, p-value = 0.03017
##
## Model df: 0.   Total lags used: 24
```

- This is a potential model; it is an optimal forecasting tool for the Solar Radiation series. Achieving the lowest MASE score of 0.2356, it demonstrates superior accuracy compared to all other models tested.
- Crucially, the model is also statistically valid: the Ljung-Box test resulted in a high p-value, and the residual plots show no significant autocorrelation (white noise). This validity means the model has captured all systematic patterns (a highly stable, proportional seasonal cycle and a negligible trend), ensuring its forecasts are reliable for future predictions.

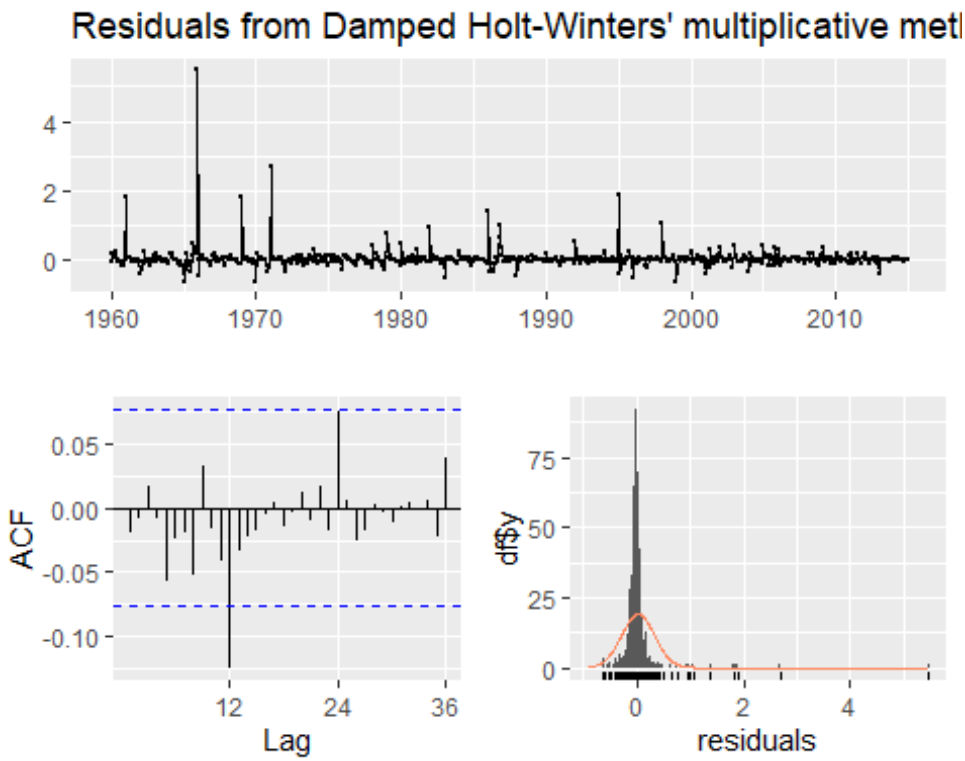
```r
summary(fit4.mul.damp)
```

```
##
## Forecast method: Damped Holt-Winters' multiplicative method
##
## Model Information:
## Damped Holt-Winters' multiplicative method
##
```

```
## Call:
## hw(y = solar_ts, seasonal = "multiplicative", damped = TRUE)
##
##   Smoothing parameters:
##     alpha = 0.805
##     beta  = 1e-04
##     gamma = 0.0124
##     phi   = 0.94
##
##   Initial states:
##     l = 9.9033
##     b = 0.4703
##     s = 0.4341 0.5649 0.8263 1.1716 1.4514 1.6316
##            1.5503 1.3834 1.1045 0.8837 0.5761 0.4221
##
##   sigma:  0.3145
##
##      AIC     AICc      BIC
## 6366.701 6367.768 6447.561
##
## Error measures:
##                      ME     RMSE      MAE       MPE     MAPE      MASE
## Training set -0.0456208 2.045442 1.239102 -2.172476 10.00296 0.2035619
##                    ACF1
## Training set 0.04177493
##
## Forecasts:
##          Point Forecast       Lo 80     Hi 80       Lo 95     Hi 95
## Jan 2015       5.611616   3.3502090  7.873023   2.1530925  9.070139
## Feb 2015       7.060438   3.3373603 10.783517   1.3664818 12.754395
## Mar 2015      10.515176   3.8550902 17.175263   0.3294535 20.700899
## Apr 2015      12.993446   3.5145037 22.472389  -1.5033454 27.490238
## May 2015      16.284381   2.9386340 29.630128  -4.1261776 36.694939
## Jun 2015      18.261020   1.7242212 34.797820  -7.0298315 43.551872
## Jul 2015      19.031597   0.2101276 37.853067  -9.7533565 47.816551
## Aug 2015      17.105654  -1.2074754 35.418784 -10.9018606 45.113169
## Sep 2015      13.764743  -2.0801991 29.609686 -10.4680049 37.997491
## Oct 2015       9.774733  -2.2585804 21.808047  -8.6286319 28.178099
## Nov 2015       6.581938  -2.0456267 15.209503  -6.6127835 19.776659
## Dec 2015       5.163049  -2.0168711 12.342970  -5.8176914 16.143790
## Jan 2016       5.616270  -2.6571166 13.889656  -7.0367828 18.269322
## Feb 2016       7.066360  -3.9144360 18.047156  -9.7273184 23.860039
## Mar 2016      10.524088  -6.6891829 27.737360 -15.8013383 36.849515
## Apr 2016      13.004566  -9.3405081 35.349641 -21.1692761 47.178409
## May 2016      16.298445 -13.0720043 45.668893 -28.6197807 61.216670
## Jun 2016      18.276925 -16.2140342 52.767885 -34.4724452 71.026296
## Jul 2016      19.048304 -18.5466197 56.643228 -38.4481704 76.544779
## Aug 2016      17.120781 -18.1782620 52.419825 -36.8644468 71.106009
## Sep 2016      13.777000 -15.8651593 43.419159 -31.5567705 59.110770
## Oct 2016       9.783493 -12.1627698 31.729756 -23.7804195 43.347405
```

```
## Nov 2016       6.587872  -8.8063951 21.982139 -16.9556278 30.131371
## Dec 2016       5.167730  -7.4021686 17.737629 -14.0562710 24.391731
```

```
checkresiduals(fit4.mul.damp)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Damped Holt-Winters' multiplicative method
## Q* = 23.57, df = 24, p-value = 0.4864
##
## Model df: 0.   Total lags used: 24
```

4. State-space models

- The analysis proceeded with a comprehensive evaluation of the State-Space (ETS) model family, which mathematically generalizes the Holt-Winters exponential smoothing technique. This approach systematically tests different combinations of Error type, Trend type, and Seasonal type to identify the structure that best fits the data
- From the above Holt-Winters models, we have 8 state-space models. A loop was created to compare the AIC, BIC, and MASE of each model.
- Within the ETS framework, the three models achieved the lowest AIC and mase are AAdA, AAA, and MMdM, respectively.

```
# Define model codes (use "_d" to indicate damped)
models <- c("AAA", "MAA", "AAA_d", "MAA_d", "MAM", "MMM", "MAM_d", "MMM_d")
```

```r
# Fit all models and collect metrics in one line
ets_results <- do.call(rbind, lapply(models, function(m) {
  damped_flag <- grepl("_d$", m)
  model_code <- gsub("_d$", "", m)
  fit <- ets(solar_ts, model=model_code, damped=damped_flag)
  data.frame(
    Model = m,
    AIC   = fit$aic,
    BIC   = fit$bic,
    MASE  = accuracy(fit)[1, "MASE"]
  )
}))

# Sort by MASE
ets_results_sorted <- ets_results[order(ets_results$MASE), ]
ets_results_sorted
```

```
##   Model      AIC      BIC      MASE
## 3 AAA_d 5428.422 5509.282 0.2461797
## 1   AAA 5434.708 5511.076 0.2471600
## 8 MMM_d 5995.550 6076.410 0.3201193
## 7 MAM_d 5953.502 6034.363 0.3222574
## 5   MAM 6105.959 6182.327 0.3721664
## 4 MAA_d 6469.079 6549.940 0.3798095
## 2   MAA 7602.755 7679.123 0.4748561
## 6   MMM 6670.168 6746.536 0.5292151
```

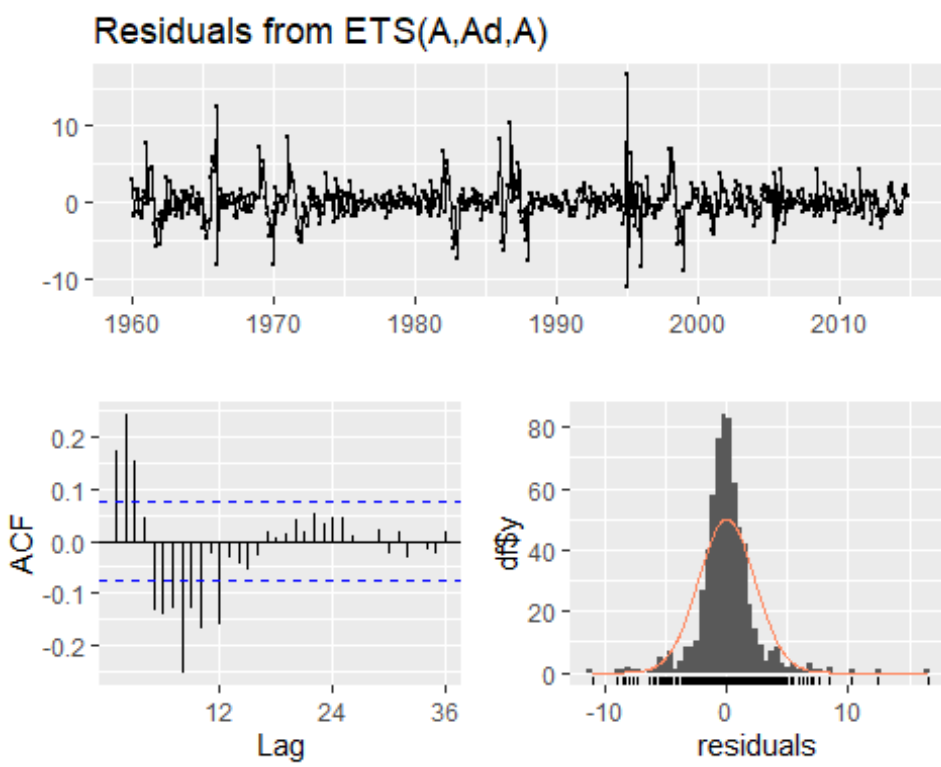
- The ETS(A,Ad,A) model (Additive Error, Additive Damped Trend, Additive Seasonality) was automatically identified as the best-fitting model within the ETS family based on its MASE score. However, the critical residual checks show that it's not suitable for reliable forecasting.
- The failure of the Ljung-Box test is due to strong, unmodeled patterns left in the errors. The ACF plot for this model shows a large, statistically significant spike at Lag 1 and several large spikes near Lag 12, indicating that the Additive formulation of the seasonal component was insufficient to capture the true, likely multiplicative, seasonal behavior of the data.

```r
ets.AAdA <- ets(solar_ts, model = "AAA", damped = TRUE)
summary(ets.AAdA)
```

```
## ETS(A,Ad,A)
##
## Call:
## ets(y = solar_ts, model = "AAA", damped = TRUE)
##
##   Smoothing parameters:
##     alpha = 0.9999
```

```
##     beta  = 1e-04
##     gamma = 1e-04
##     phi   = 0.9388
##
##   Initial states:
##     l = 11.154
##     b = 0.7632
##     s = -10.4919 -8.137 -3.348 2.5794 8.08 11.1219
##            9.9586 6.9916 1.9573 -1.8565 -7.1607 -9.6946
##
##   sigma:  2.3446
##
##      AIC     AICc      BIC
## 5428.422 5429.489 5509.282
##
## Training set error measures:
##                        ME     RMSE      MAE       MPE     MAPE      MASE
## Training set -0.01091357 2.314163 1.498521 -1.468083 12.44796 0.2461797
##                   ACF1
## Training set 0.1700724
```

```
checkresiduals(ets.AAdA)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,Ad,A)
```

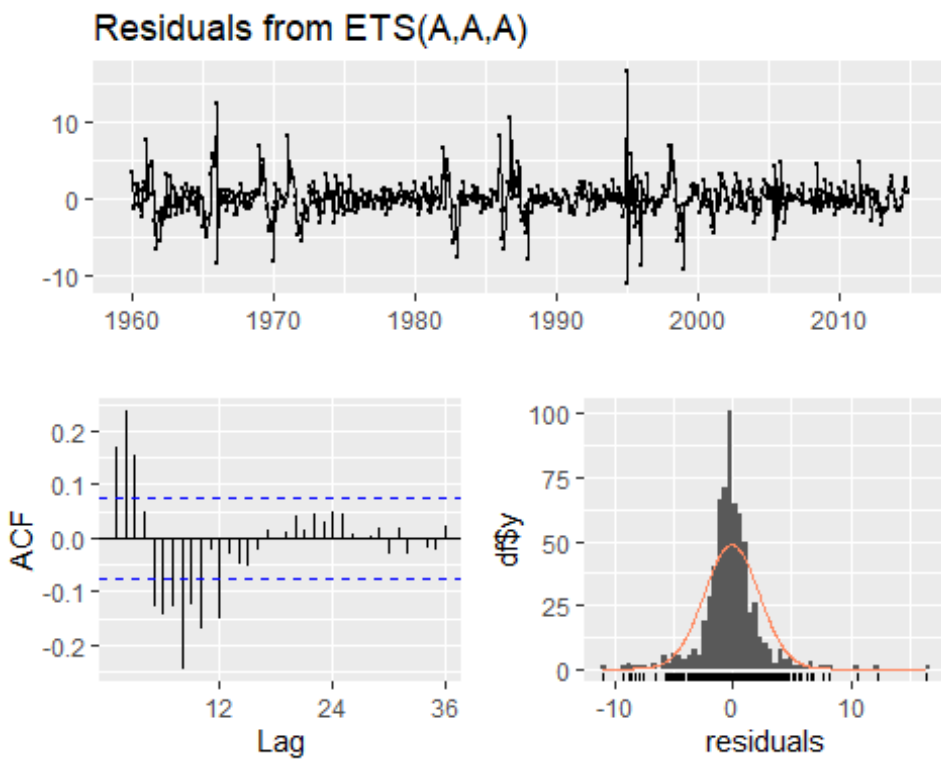
```
## Q* = 210.76, df = 24, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 24
```

- This model, which represents the standard undamped Holt-Winters additive method within the ETS framework, performed slightly worse than its damped counterpart (ETS(A,Ad,A)), and both are statistically flawed.
- Both ETS models are statistically invalid, despite their competitive MASE scores

```r
ets.AAA <- ets(solar_ts, model = "AAA", damped = FALSE)
summary(ets.AAA)
```

```
## ETS(A,A,A)
##
## Call:
## ets(y = solar_ts, model = "AAA", damped = FALSE)
##
##   Smoothing parameters:
##     alpha = 0.9993
##     beta  = 0.0019
##     gamma = 1e-04
##
##   Initial states:
##     l = 11.3306
##     b = 0.209
##     s = -10.7542 -8.4968 -3.227 3.0768 7.9264 10.9122
##            10.1422 7.3322 2.2353 -1.9528 -7.3626 -9.8316
##
##   sigma:  2.3575
##
##      AIC     AICc      BIC
## 5434.708 5435.662 5511.076
##
## Training set error measures:
##                      ME     RMSE      MAE       MPE     MAPE    MASE
ACF1
## Training set -0.1179476 2.328736 1.504489 -2.230359 12.68101 0.24716
0.1703355
```

```r
checkresiduals(ets.AAA)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,A,A)
## Q* = 205.55, df = 24, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 24
```

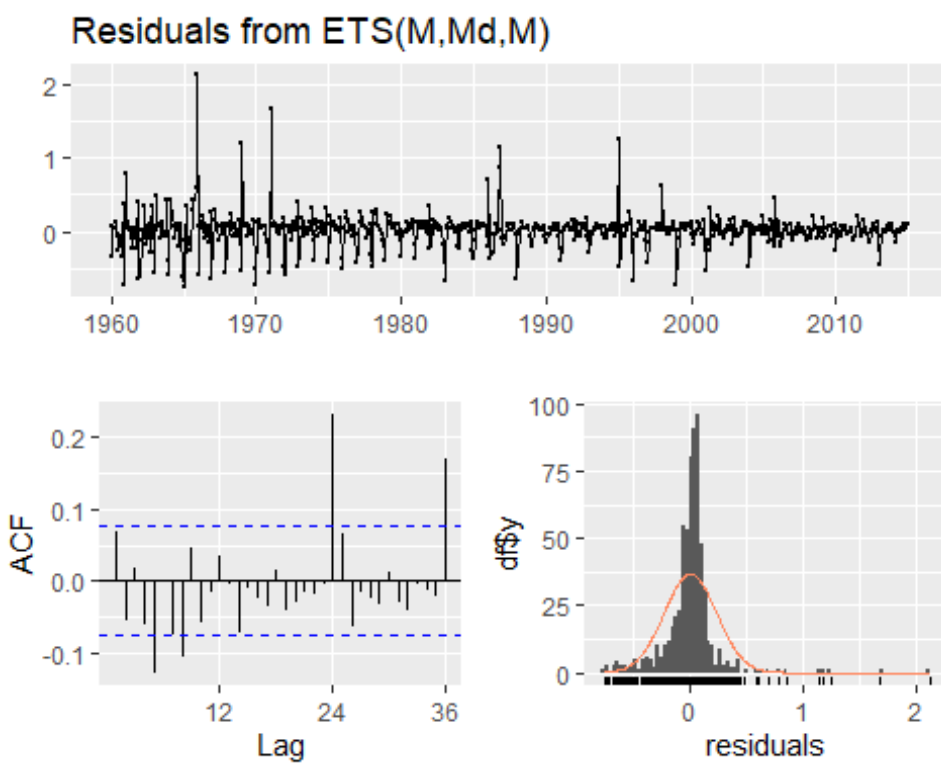
- This model, the multiplicative counterpart to the previously failed additive models, was the third-best performing model in your automated ETS search. The ACF plot for ETS(M,Md,M) shows a significant spike at Lag 24 (and near Lag 1), indicating that residual correlation remains in the errors. This failure of the automated ETS fit to fully smooth the errors is why its p-value is so low and why it cannot be trusted for long-term forecasting.

```r
ets.MMdM <- ets(solar_ts, model = "MMM", damped = TRUE)
summary(ets.MMdM)
```

```
## ETS(M,Md,M)
##
## Call:
## ets(y = solar_ts, model = "MMM", damped = TRUE)
##
##   Smoothing parameters:
##     alpha = 0.7892
##     beta  = 0.016
```

```
##     gamma = 0.0696
##     phi   = 0.9575
##
##   Initial states:
##     l = 9.2782
##     b = 1.1745
##     s = 0.7133 0.3063 0.6852 1.1282 1.2133 1.5082
##            1.7211 1.2962 1.1358 0.9309 0.656 0.7056
##
##   sigma:  0.2341
##
##      AIC     AICc      BIC
## 5995.550 5996.617 6076.410
##
## Training set error measures:
##                       ME     RMSE      MAE       MPE     MAPE      MASE
## Training set 0.02448728 3.076938 1.948599 -5.645612 16.87559 0.3201193
##                    ACF1
## Training set 0.1452617
```

```
checkresiduals(ets.MMdM)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(M,Md,M)
## Q* = 79.153, df = 24, p-value = 8.298e-08
```

```
##
## Model df: 0.   Total lags used: 24
```

- The final ranking confirms the Holt-Winters Multiplicative Damped model as the optimal forecasting tool. It achieved the lowest MASE of 0.20356, demonstrating a significant improvement in accuracy over the best ARDL model (MASE 0.4479). Crucially, this model was the only top-performing candidate that was also statistically valid, successfully passing the Ljung-Box test and ensuring the reliability of its forecasts. All other top exponential smoothing models (AAA and AAA_d) failed the residual tests, reinforcing the initial choice.

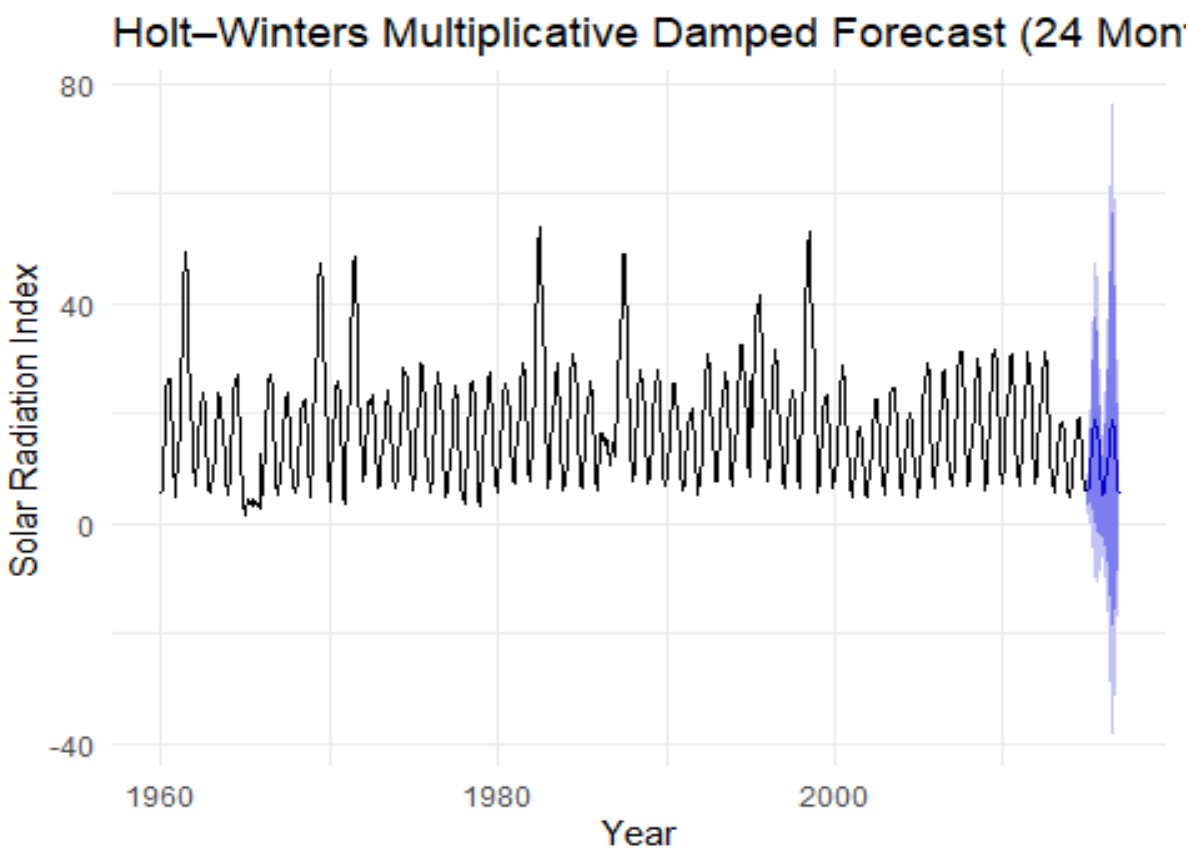
Comparative board

```
sp_results <- data.frame( #comparative board for state-space models
  Model = c("AAdA", "AAA", "MMdM"),
  AIC = c(ets.AAdA$aic, ets.AAA$aic, ets.MMdM$aic),
  BIC = c(ets.AAdA$bic, ets.AAA$bic, ets.MMdM$bic),
  MASE = c(accuracy(ets.AAdA)[,"MASE"][1], accuracy(ets.AAA)[,"MASE"][1],
           accuracy(ets.MMdM)[,"MASE"][1]))
compared_board <- rbind(compared_board, sp_results) #combine with previous
results
compared_board <- compared_board[order(compared_board$MASE), ]
compared_board

##                          Model      AIC      BIC      MASE
## 41   HW Multiplicative Damped 6366.701 6447.561 0.2035619
## 31          HW Multiplicative 6648.746 6725.114 0.2233077
## 22         HW Additive Damped 5428.422 5509.282 0.2461797
## 13                       AAdA 5428.422 5509.282 0.2461797
## 12                HW Additive 5434.708 5511.076 0.2471600
## 23                        AAA 5434.708 5511.076 0.2471600
## 32                       MMdM 5995.550 6076.410 0.3201193
## 7             ADLM (p=5, q=5) 3097.877 3156.177 0.4479311
## 6             ADLM (p=4, q=5) 3096.024 3149.839 0.4479481
## 5             ADLM (p=2, q=5) 3097.288 3142.134 0.4503599
## 4                       Koyck 3946.476 3964.439 1.0324829
## 1                  Finite DLM 4578.787 4645.895 1.5516004
## 3  Polynomial DLM (q=12, k=2) 4567.969 4590.339 1.5630351
## 2  Polynomial DLM (q=11, k=2) 4577.156 4599.533 1.5704569
## 11           Dynamic Linear 1 2861.980 2969.611        NA
## 21           Dynamic Linear 2 2863.925 2976.041        NA
```

Forecasting

- The plot demonstrates a statistically sound forecast that respects the strong seasonality and damping observed in the solar radiation data, aligning with the low MASE score (0.20356)
- However, the shaded prediction intervals get progressively wider as the forecast moves further into the future.

```r
fit_hw_md <- hw(solar_ts, seasonal = "multiplicative", damped = TRUE, h = 24)
#forecast 2 years ahead
autoplot(fit_hw_md) +
  labs(
    title = "Holt-Winters Multiplicative Damped Forecast (24 Months Ahead)",
    y = "Solar Radiation Index",
    x = "Year"
  ) +
  # Customize colors/theme
  theme_minimal() +
  scale_color_manual(values = c("Data" = "darkgray"))
```



Task 2: Analyse correlation

Data Description

- The analysis began by importing the dataset and converting the relevant financial variables into quarterly time series objects, which were initialized to start in the third quarter of 2003.

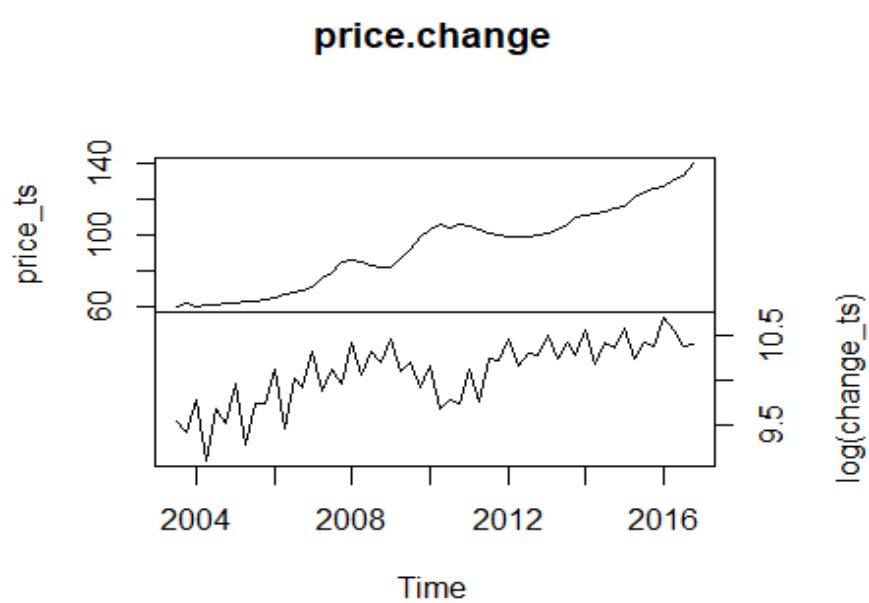
```r
# read data
data2<-read.csv("data2.csv")
```

```r
# examine
head(data2)

##    Quarter price change
## 1 Sep-2003  60.7  14017
## 2 Dec-2003  62.1  12350
## 3 Mar-2004  60.8  17894
## 4 Jun-2004  60.9   9079
## 5 Sep-2004  60.9  16210
## 6 Dec-2004  62.4  13788

# time series object
price_ts <- ts(data2$price, start = c(2003, 3), frequency = 4)
change_ts <- ts(data2$change, start = c(2003, 3), frequency = 4)
data2_ts <- ts(data2 [,2:3], start = c(2003, 3), frequency = 4)
```

Correlation check

- Through the plot, the two series appear to move together visually. The visual co-movement supports the high correlation but does not prove a causal or genuine short-run relationship.
- The correlation coefficient indicates a strong positive contemporaneous linear relationship. However, it is highly likely to be spurious:
    - The two series exhibit a non-stationary pattern, and any two non-stationary series that trend in the same direction will naturally produce a high correlation coefficient.

```r
#plot the correlation
price.change = ts.intersect(price_ts,log(change_ts))
plot(price.change,yax.flip=T)
```

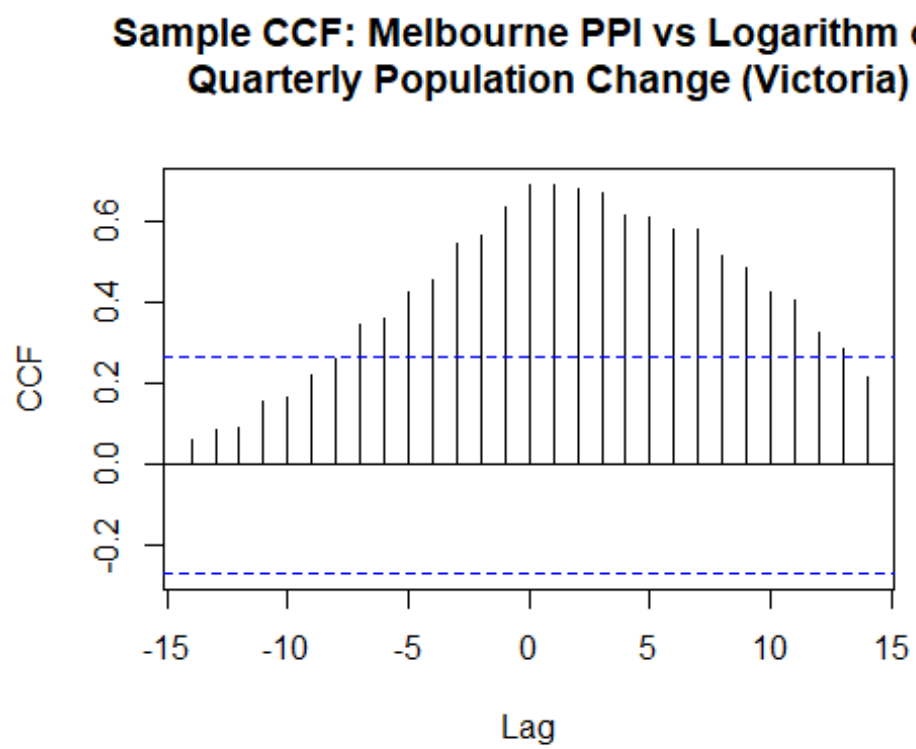


```r
# correlation coefficient
cor(price_ts,change_ts)

## [1] 0.6970439
```

- The high, slow-tapering correlations across many lags confirm that the measured relationship is driven by the shared non-stationary behavior (trend) of the series
- The stationarity will be tested for both series to prove or disprove spuriousness

.

```r
ccf(as.vector(price.change[,1]), as.vector(price.change[,2 #CCF plot
 ]),ylab='CCF', main = "Sample CCF: Melbourne PPI vs Logarithm of
    Quarterly Population Change (Victoria)")
```

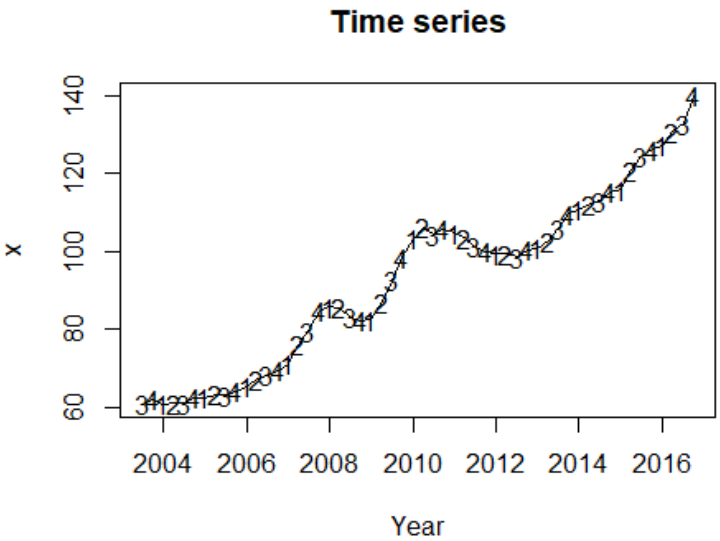


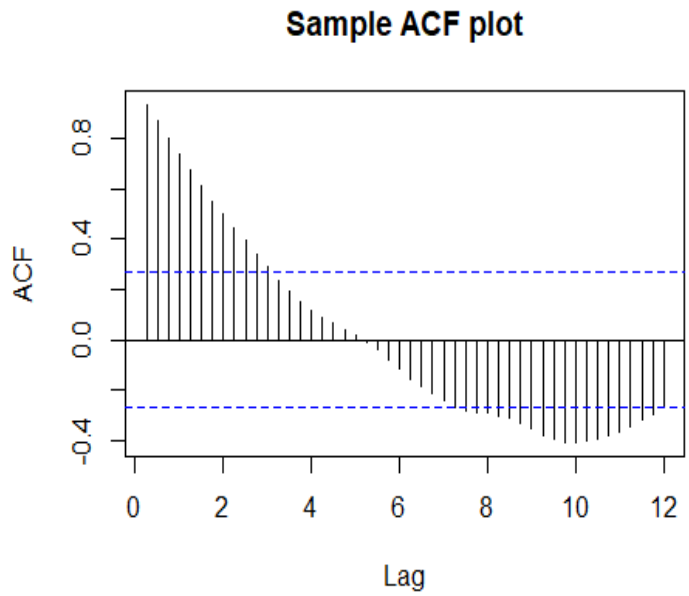
Non-stationarity examination

- The plot clearly shows a strong upward trend in the Residential Property Price Index (PPI) over the entire period, confirming our earlier visual assessment.
- The ACF shows a slowly decaying pattern, where the autocorrelation coefficients remain high and significant for many lags before eventually dropping. This persistence is the classic visual signature of a non-stationary series with a trend (Kis, 2024).
- The P-value of 0.8458 is much greater than the standard significance level of 0.05; we fail to reject the null hypothesis, and the series is non-stationary.

```r
plots <- function(x) { #function to explore time series objects: time series plot, ACF plot and ADF test
  plot(x, xlab='Year', main="Time series")
  points(y=x,x=time(x), pch=as.vector(season(x)))
  acf(x, main="Sample ACF plot", lag.max = 48) #ACF plot
  adf <-adf.test(x)
  print(adf)
}
```

```r
plots(price.change[,1])
```

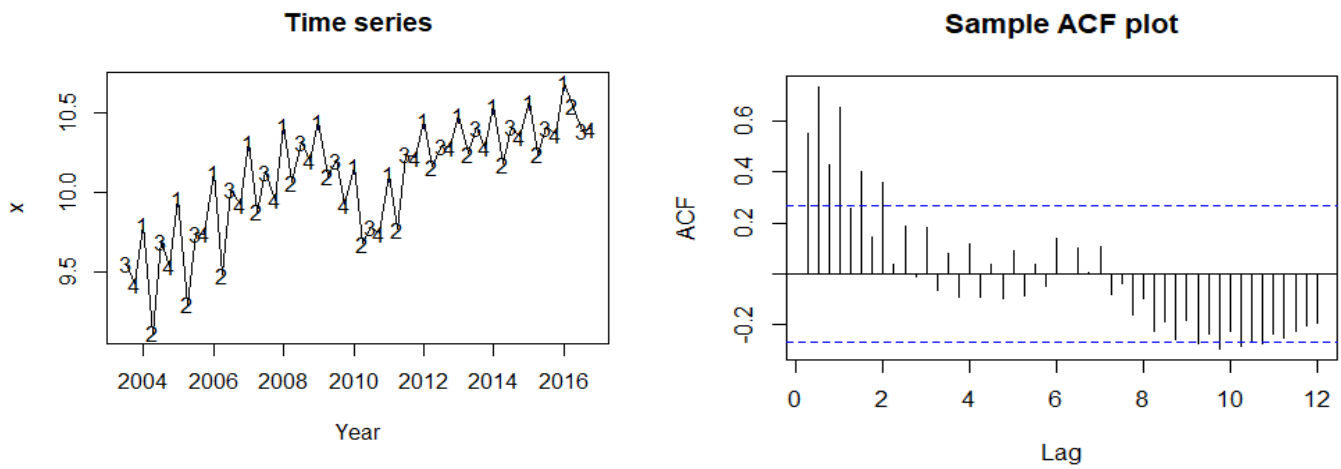





```
##
##  Augmented Dickey-Fuller Test
##
## data:  x
## Dickey-Fuller = -1.3264, Lag order = 3, p-value = 0.8458
## alternative hypothesis: stationary
```

- The logarithm of quarterly population change has both a trend component and a strong seasonal component
- The ACF exhibits a slow decay, with positive and significant autocorrelation coefficients lasting for multiple lags.
- The P-value is 0.6077, much higher than 0.05, so we fail to reject the null hypothesis. This series is non-stationary

```r
plots(price.change[,2])
```




```
##
##  Augmented Dickey-Fuller Test
##
## data:  x
## Dickey-Fuller = -1.9176, Lag order = 3, p-value = 0.6077
## alternative hypothesis: stationary
```

Differencing

- The seasonal differencing is applied at lag 4, followed by first differencing for both series. This approach is standard for data that is both non-stationary in trend and seasonal.

```r
price_diff <- diff(diff(price.change[,1]),4) #first differencing and seasonal differencing
adf.test(price_diff)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  price_diff
## Dickey-Fuller = -5.1122, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary
```

```r
change_diff <-diff(diff(price.change[,2],4))
adf.test(change_diff)
```

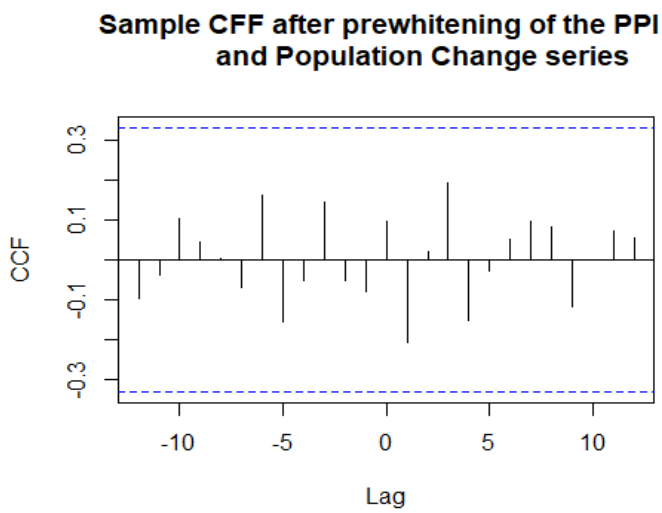
```
##
##  Augmented Dickey-Fuller Test
##
## data:  change_diff
```

```
## Dickey-Fuller = -3.7116, Lag order = 3, p-value = 0.03326
## alternative hypothesis: stationary
```

Prewhitening and Conclusion

- The strong initial correlation between the Melbourne Property Price Index and Victoria's Quarterly Population Change is spurious. Once the non-stationary trends and seasonality are removed through differencing and prewhitening, there is no statistically significant evidence of a short-run linear relationship between the two series.

```
me.dif=ts.intersect(price_diff,change_diff)
prewhiten(as.vector(me.dif[,1]),as.vector(me.dif[,2]),ylab='CCF',
main="Sample CFF after prewhitening of the PPI
          and Population Change series") #prewhitening
```



Conclusion

Task 1: The comprehensive evaluation across all mandated model classes concluded that the Holt-Winters Multiplicative Damped exponential smoothing model yielded the superior forecast. Initial analysis showed the solar radiation series to be stationary but highly seasonal (S=12), with a strong inverse relationship to precipitation. The Holt-Winters Multiplicative Damped model successfully captured the multiplicative seasonal structure and damped the long-term trend, resulting in the lowest overall forecast error with a MASE of 0.20356. This model is therefore selected as the most accurate for the 2-year forecast.

Task 2: The comprehensive time series analysis decisively demonstrates that the strong initial correlation of 0.697 between the Melbourne Property Price Index and the Logarithm of Quarterly Population Change is spurious. Formal Augmented Dickey-Fuller (ADF) tests confirmed that both the PPI and the population change series were non-stationary in both trend and seasonality.

Reference

Kis, A. (2024, July 26). *Understanding Autocorrelation and Partial Autocorrelation Functions (ACF and PACF)*. Medium. https://medium.com/@kis.andras.nandor/understanding-autocorrelation-and-partial-autocorrelation-functions-acf-and-pacf-2998e7e1bcb5